A pre-registered test of training on co-occurring versus isolated stressors in a multi-agent fire environment

A pre-registered test of whether training a single policy on co-occurring stressors beats training it on each in isolation, in the Compound Hostile Environment (CHE), a multi-agent environment where a fire burns in every episode.

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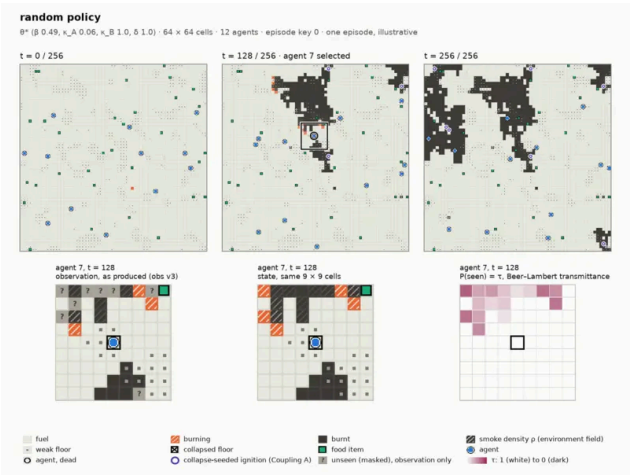
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We report a pre-registered test of whether training one policy on co-occurring stressors beats training it on each in isolation, in the Compound Hostile Environment (CHE), a pure-JAX multi-agent environment where a cellular-automaton fire burns in every episode. [1] Two stressors are coupled to the fire: structural collapse can add ignitions, and smoke attenuates perception by Beer–Lambert transmittance; the third, communication denial, is independent. Both arms train at matched compute and are evaluated at a held-out near-critical severity, with task completion and survival as Šidák-corrected co-primaries. We ran 72 seeds per arm, above the registered cap of 60, a recorded deviation made before unblinding.

Joint training raises survival by 0.96 points (Šidák-corrected 95 % CI [0.47, 1.44]); completion shows no detectable change (−1.73 points, CI [−4.12, +0.66]), which excludes a joint-training gain larger than 0.66 points. Both differences are sign unstable over the final half of training: the survival gap opens only in the last fifth, and completion leans negative from 70 % of the budget. The survival gain does not survive the inert-share correction. Communication denial is inert under a frozen random-projection encoder at this scale, so the third of the isolated arm's training episodes that carry only it are fire-only. This is our design error: we kept a stressor already found inert and did not count the fire-only share it created; found before the main training runs, the bias was registered with its correction while they ran, before unblinding. The correction is a 20-seed sensitivity arm without those episodes, outside the confirmatory family; joint training's survival differs from that arm's by −0.03 points (CI [−0.68, +0.63]), though it still sees each coupling twice as often, always combined.

The fire is an ambient hazard that no reward, cost or constraint channel reads; the reward reads only task variables and agent deaths, pricing every death the same. Severity is set relative to a measured percolation threshold, and each stressor is a parameter of one transition or observation kernel, so ablations are bitwise-exact nested models. We describe a measurement protocol with reproducibility floors per metric, per GPU and per artifact. We recommend, from this single environment, that studies of compositional generalization match the share of training with an active stressor before attributing a gain to composition.

Figure 1. One episode of CHE. The map is environment state: fire, the smoke field ρ, weak and collapsed floor, food, agents alive and dead, and ignitions seeded by collapse. Perception is only in the selected agent's three panels: its observation as produced (labelled obs v3, the masking observation), the true state of those cells, and P(seen) = τ, the transmittance through smoke. Configuration, episode key, random policy and stochastic actions were fixed before rendering. Status: illustrative · random policy · grades nothing · Interact: play, scrub, jump to events, toggle layers, select an agent.



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Each caption carries a status: illustrative (grades nothing); registered (read by a rule fixed before unblinding); sensitivity arm (outside the two confirmatory tests); secondary (carries no verdict); descriptive (no test); reference scale (cross-artifact), a spread measured on other runs, shown for scale only.

1 Introduction

When a training distribution has to cover several stressors, its designer chooses whether the policy meets them together or one at a time. Ecology studies the interactive and cumulative effects of multiple stressors [1, 2], and climate science the risk from compound events [3, 4]. In reinforcement learning the choice is a question of zero-shot generalization [5]: must a policy see

stressors co-occur in training to cope with their co-occurrence at test time? Gao et al. (2024) [6] studied it for imitation-learned manipulation; we study it for survival in multi-agent reinforcement learning.

Where the hazard sits in the learning problem shapes what such a comparison can teach: in some fire-related work it is the task [7], and safe reinforcement learning typically places it in a cost or a constraint [8]. In the Compound Hostile Environment (CHE) the fire is an *ambient hazard* (Definition 2 in §3): a fixed stochastic process that optimizes against no one and enters no reward, cost or constraint channel. The reward reads task variables and agent deaths, and prices every death at $d_p = 0.5$, whatever its cause. No reward or cost signal mentions the hazard in either arm, so the arms differ only in what their training distributions show the agents. Proposition 1 shows that the fire, which looks like nonstationarity to any one agent, is part of the state: CHE is a well-defined Dec-POMDP whose states reachable within the horizon form a finite set (§3).

CHE is written in pure JAX, and a cellular-automaton fire burns in every episode (Figure 1). For the idealized kernel the burnt region is exactly a bond-percolation cluster (Proposition 2) [9, 10], so the three severities place the fire's transmissibility β below, near and above a *measured* percolation threshold of the implemented kernel (§4). Two stressors are coupled to the fire: collapse can add ignitions (Coupling A, of strength κ_A; §5), and smoke attenuates each agent's egocentric observation by Beer–Lambert transmittance (Coupling B, κ_B; §6). A third, link denial at level δ, is independent of it (§7); $\theta = (\beta, \kappa_A, \kappa_B, \delta)$ is the environment configuration.

The training question becomes a pre-registered test of two training mixtures (Figure 2).

Figure 2. What each arm trained on. Each bar splits one arm's training mixture into episode types: Coupling A only, Coupling B only, both together, and fire-only. Beneath each bar are each coupling's frequency (labelled "per-stressor marginal" on the figure) and the active share. ISO-4 is ISO without its δ-only components. The figure shows configured shares, read from the training configs, and cannot show outcomes, trained behaviour or predictions. Status: configuration facts · grades nothing · Interact: count δ-only episodes as inert (default; §7) or nominally; split by β; hover a segment; table view.

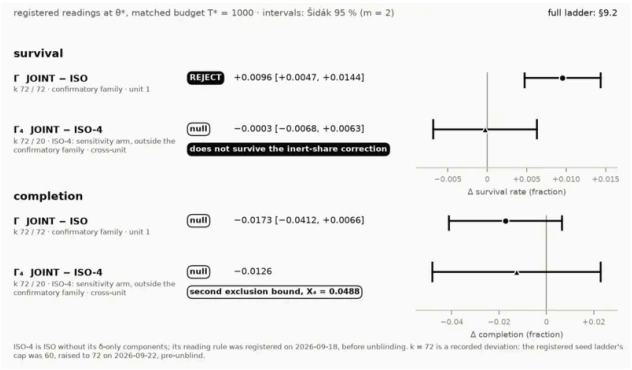


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An isolated arm (ISO) trains one policy on single-stressor configurations and a joint arm (JOINT) one on configurations with every stressor on, both at β = 0.43 and 0.70. Both are evaluated at θ*, every stressor on at the near-critical β = 0.49, which neither saw. Architecture, hyperparameters, evaluation draw and compute (1000 updates) are matched, with one shared-parameter IPPO policy per training seed [11]. The estimand is the compositional gap Γ, JOINT's mean minus ISO's at θ*, on two co-primary outcomes, task completion and survival, Šidák-corrected for m = 2 tests. Four outcome branches, a framing for each and a falsifier were registered on 2026-08-13, before the grid began (a fifth was added before unblinding; Appendix E). We ran 72 seeds per arm, above the registered cap of 60, a recorded deviation made before unblinding (§9.1).

JOINT sees each coupling three times as often as ISO, so the endpoints differ in frequency as well as in co-occurrence. A secondary sweep in the spirit of DBCA [12] varies co-occurrence at fixed frequency, but its fire-only share rises with co-occurrence, and it bears no verdict (§9.1). Communication denial is inert under our frozen message encoder at this scale (§7), so the composition tested is that of the two couplings, and the third of ISO's training episodes that carry only δ are fire-only. ISO's *active share*, the fraction of training episodes with a behaviourally active stressor, is therefore two thirds, and JOINT's is one. This is an error in our design: it left δ out of the secondary sweeps because δ was inert, yet kept it in ISO's mixture, and did not count the fire-only episodes that resulted. A review of the registered design identified the bias on 2026-08-16, before the grid; we registered it, its direction, and a correcting arm, ISO-4, with its reading rule on 2026-09-18, mid-grid and before unblinding (§9.1). At the test's severity, Coupling B's own survival effect in calibration was within the spread over three seeds (§10).

Figure 3. Survival: Γ REJECT; Γ_4 null, "does not survive the inert-share correction". Completion: Γ null; Γ_4 null, a second exclusion bound. Γ is JOINT − ISO, the registered test; Γ_4, the corrected gap, is JOINT − ISO-4, where ISO-4 is a sensitivity arm outside the confirmatory family. X_4 is Γ_4's minimum detectable effect at 80 % power. The text quotes these rate differences × 100, as points. On the figure "unit 1" is GPU 1, "cross-unit" means across GPUs, and "full ladder: §9.2" points to Figure 11. Γ_4's completion interval is drawn but not printed. The figure cannot show how Γ moved during training (sign unstable over the final half; Figure 12). Status: registered for Γ · sensitivity arm for Γ_4 · Interact: none.



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At θ*, joint training raises survival by 0.96 points (Šidák-corrected 95 % CI [+0.47, +1.44]; points are rate differences × 100), and the survival test rejects (Figure 3). Task completion shows no detectable change, −1.73 points (CI [−4.12, +0.66]), which excludes a joint-training gain larger than 0.66 points. The founding hypothesis, that joint training improves both outcomes, did not occur. The falsifier registered for an asymmetric outcome passes (§9.2); under Definition 2 the asymmetry lies between a penalized channel, deaths, and a rewarded one, the task. Both co-primaries are sign unstable over the final half of training: the survival gap opens only in the last fifth of the budget, and completion leans negative from 70 % of it onward (Figure 12).

The survival gain does not survive the inert-share correction. ISO-4, ISO without its δ-only components, is a sensitivity arm outside the confirmatory family, trained with 20 seeds on a second GPU; its survival is 0.98 points above ISO's, an estimate with no registered test. The corrected gap Γ_4, JOINT − ISO-4, is −0.03 points (Šidák-corrected 95 % CI [−0.68, +0.63]), and a same-GPU replicate, proposed after Γ_4 was seen, agrees. The correction excludes only a large composition effect, and ISO-4 changes the active share and each coupling's frequency together; §9.2 states both limits, what the secondary design adds and where its fit fails. Once the active share is matched, JOINT's remaining differences add nothing detectable to survival, although it still sees each coupling twice as often as ISO-4 and always in combination.

A joint-versus-isolated comparison that leaves the active share free can report an active-share effect as a composition effect (§9.3). In the search described in §2 we found no reinforcement-learning comparison that matches its arms on that share. We claim no convergence guarantee, no survival gap robust to the training budget (every comparison is at matched compute), and no theorem that joint training wins: Proposition 4 (§9.1) only bounds what isolated training can be guaranteed to transfer.

Our contributions are:

1. CHE, an environment built as an instrument, with severity by measured dynamical phase, bitwise-exact nested ablations and theory that doubles as unit tests (§3–§7).
2. Calibration findings that stand without the training comparison. Under the calibration's trained policies, collapse-seeded ignitions fall 4.8-fold from low to high severity, a fall the random-policy calibration had already shown before any trained policy was measured (§5). Smoke costs survival at high severity, in the same direction in three of three replicates (§6), and communication denial is inert under the frozen encoder (§7). A measurement protocol gives acceptance bars floors measured per metric, per GPU and per artifact, and names the bars that lack one (§8).
3. A pre-registered test of joint against isolated training (§9), whose survival advantage does not survive the inert-share correction, registered before unblinding.

2 Related work

Table 1 sets the nearest fire-related and safety-related work beside CHE.

Table 1. Where the hazard enters the learning or planning problem.

work	agents and method	where the hazard enters
Haksar & Schwager (2018) [7]	aerial agents fighting wildfire; multi-agent deep Q-network	**the task**: a reward for moving to burning trees on the fire's boundary (their Algorithm 2); the paper gives the fire no way to harm the agents
JaxWildfire (Çakır et al., 2025) [13]	one air-tanker agent; JAX cellular-automaton wildfire simulator	**the reward**, hazard-priced: a per-step penalty proportional to the number of burning cells, with a bonus for full extinction; harm to the agent is not described
safe reinforcement learning, as García & Fernández (2015) define it [8]	learning policies that maximize return while ensuring reasonable system performance and/or respecting safety constraints	**a constraint or a cost**, typically
VULCAN (Liu & Yan, 2026) [14]	a proposed approach to multi-robot fire search and rescue, planned by a vision-language model rather than learned	**a hazard cost map** that folds in smoke, heat and sensing uncertainty

work	agents and method	where the hazard enters
CHE (this paper)	12 agents; independent PPO	**ambient**: the transition and observation kernels only; no reward, cost or constraint reads the fire, and the reward prices each death at $d_p = 0.5$, whatever its cause

CHE's reward is hazard-blind, while JaxWildfire's is hazard-priced: ours prices an outcome, each death the same (Definition 2 in §3.3). VULCAN's preliminary evaluation covers baselines only, and its evaluation metric, cumulative hazard exposure, is also abbreviated CHE, unrelated to our environment's name.

Coupling A lets structural collapse add ignitions. The nearest neighbour we found is Raut et al. (2026) [15], who model public-safety power shutoffs as a single-agent decision problem. In it, the power flows that follow from the operator's switching decisions set the failure probability of lines in wildfire areas, which the authors treat as ignition risk. Their reward is negative operating cost, and there is no spatial spread model. CHE's agents influence the hazard too, but weakly and indirectly: an agent standing on a weak cell raises that cell's collapse rate, yet most collapses happen with no agent present (§5.1). A collapse may then seed fire, so we do not claim that CHE's hazard grows without agent influence. The difference from Raut et al. is one of directness and size: there the controlled decision sets the failure probability, while in CHE ignition is not an action available to the agents.

Coupling B attenuates each agent's observation by Beer–Lambert transmittance along the line of sight through smoke. VULCAN degrades simulated sensing with smoke density and visibility loss, and infers smoke density from image degradation and depth consistency, but it has no transmittance model. JaxWildfire's observations are partial and egocentric but not attenuated. In the search described below, we found no reinforcement-learning environment whose observation kernel attenuates by transmittance.

CHE follows JaxMARL's environment conventions [16] and trains with independent PPO [11]. None of the JAX multi-agent benchmarks we surveyed, JaxMARL, Multi-Agent Craftax [17] and Assistax [18], has an evolving, spreading lethal hazard field or perception attenuated by a hazard. Multi-Agent Craftax comes closest in kind: its agents lose health to hunger and enemies and can die, but its lava is static terrain that agents cannot enter. In Crafter and in Craftax-Classic, the JAX reimplementation described alongside Craftax, lava is lethal, and Crafter and Craftax reward each health point regained and penalize each one lost [19, 20], pricing an outcome, as our reward does. POBAX [21] is a single-agent suite for partial observability, and BenchMARL [22] a multi-agent benchmarking library built on TorchRL.

Domain randomization trains on a broad, fixed distribution over simulator parameters so that the deployed setting looks like one more sample [23, 24]. Unsupervised environment design adapts the distribution to the learner instead. PAIRED generates environments from a regret objective [25]; Prioritized Level Replay revisits levels by their estimated learning potential [26] and was later recast as environment design [27]; and ACCEL evolves levels near the frontier of the agent's ability [28]. These methods ask which distribution to train on, usually adaptively; we compare two fixed distributions and evaluate no adaptive curriculum (§10).

The closest earlier study of our question is Gao et al. (2024) [6]. In visual imitation learning for robotic manipulation, they ask whether policies compose environmental factors, such as object type and position, table texture and height, and camera position, to succeed on unseen combinations. They compare data-collection strategies at a matched number of demonstrations, among them varying one factor at a time and cycling through factors so that more combinations appear (their §III-C and §IV-A). Policies do compose, more weakly when both factors are physical (their §V-B), and the strategy that covers more combinations does better. Their design is built for data efficiency and does not separate co-occurrence from how often each factor value appears; ours matches compute and, in the ISO-4 correction, the active share (§1). CompoSuite [29] studies the composition of tasks, over robots, objects, objectives and obstacles, in multi-task reinforcement learning. The DBCA method [12] builds splits whose atom distributions are similar and whose compound distributions differ. Our secondary sweep follows that idea, holding each coupling's frequency fixed while co-occurrence varies; our ISO and JOINT endpoints do not, since JOINT sees each coupling three times as often as ISO (§1, §9.1).

Robust reinforcement learning asks the same question about perturbations. Erdem & Üre (2025) [30] train single-agent continuous-control policies against a learned adversary that perturbs observations, actions, or both, and report that agents trained against one modality remain vulnerable to simultaneous attacks on both (their Tables 3–4). Under single-type attacks, agents trained against their balanced mixed adversary do at least as well as those trained against a naive one in all six cells, one a tie (their Table 5). Of their six training arms, one is an unattacked baseline, and in each of the others an attack is computed at every step, with no scheduled clean steps (their Algorithm 1). No arm has a counterpart of the fire-only share our correction removes. No arm trains on a mixture of single-type attacks, and the mixed arms carry the combined budget, which the authors note is more severe. Their design thus bundles per-modality frequency, co-occurrence and a larger budget, which is nearer to what our corrected contrast leaves in, where JOINT still sees each coupling twice as often as ISO-4; it neither supports nor contradicts our corrected null. Agrawal et al. (2023) [31] train multi-agent policies against random reward, state and action noise, one or two types at a time, under a curriculum that raises the noise level and sets training length, so exposure differs between their models. In supervised vision, Tramèr & Boneh (2019) [32] and Hsiung et al. (2023) [33] study robustness to multiple and to composite perturbations.

Chen et al. (2022) [34] model domain randomization as returning the optimal history-dependent policy of a latent MDP, a set of fully observed MDPs that differ in their transitions. Every upper bound they prove on its shortfall assumes each MDP is communicating, with bounded diameter. Theorem 1 (§6) is kin to the construction of their Proposition 1, in which an irreversible choice is made before anything distinguishes the candidate MDPs. Read as a latent MDP with the signals encoded as states, Theorem 1's two-corridor toy has an absorbing failure, so their upper bounds do not apply. It also measures a different gap, the adaptive optimum against the policy optimal for a fixed map. Chen et al. therefore do not subsume Theorem 1, and Theorem 1 does not extend their bounds. Pynadath & Tambe (2002) [35] relate the complexity of team decision problems with communication to observability. Free communication lowers the complexity when the world is collectively observable (their Theorem 4 and Table 2), and makes no difference when it is individually observable or not observable at all (their Theorems 5–6). Remark 2' (§7) refines this for collective partial observability: communication's marginal value vanishes when interchangeable agents' views are redundant, and returns when roles are irreplaceable or deaths are costly. A survey of adaptability in multi-agent reinforcement learning (Hu et al., 2025, revised 2026) [36] calls an environment family highly diagnostic if it "systematically varies" factors such as population, observability or communication structure "while holding other factors fixed". It mentions compositional generalization but does not analyse shifts in several factors at once.

We ran the literature search with the help of LLM-based research assistants, between August and October 2026. A census of benchmark venues and a search for theorems that could subsume ours came first; a search of domain randomization, environment design, multi-task and robust-learning neighbours followed. Both used web search and the arXiv API and read abstracts. The works this section describes in detail were then read in full; the remaining citations rest on their abstracts. We did not query IEEE Xplore. Wherever this paper says that no work does something, it means no work found by this search.

3 The environment

CHE is a grid world in which a team of agents forages while a fire burns; Figure 4 shows one step.

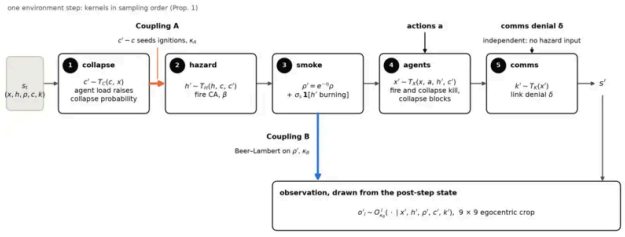


Figure 4. One environment step of CHE.

Collapse, fire, smoke, agents and communication are sampled in the order of Proposition 1. Coupling A (orange) carries c' − c into the fire; Coupling B (blue) carries ρ' into the observation, drawn from s'. The comms kernel takes δ and reads no hazard; T_X reads the post-step h' and c', as in the code. Arrow colours are this figure's own. The reward is not drawn (Definition 2). Status: schematic, no data.

3.1 State, kernel and step order

Definition 1 (Compound Hostile Environment). A CHE is a Dec-POMDP $\mathcal{M}_\theta = \langle I, \mathcal{S}, \{\mathcal{A}_i\}, T_\theta, R, \{\Omega_i\}, O_\theta, \gamma \rangle$ with a finite set of agents I on a grid G, action sets $\mathcal{A}_i$, observation sets Ω_i,

reward R (Definition 2), observation kernel O_θ and discount γ. It is indexed by the *environment configuration* $\theta = (\beta, \kappa_A, \kappa_B, \delta)$: the fire's per-neighbour ignition probability β, the strengths κ_A and κ_B of Couplings A and B, and the communication-denial level δ. Its state space factors as

$$\mathcal{S} = \mathcal{X} \times \mathcal{H} \times \mathcal{P} \times \mathcal{C} \times \mathcal{K} \times \{0, \dots, T\},$$

where $x \in \mathcal{X}$ holds the agents' positions $x_i \in G$, their alive flags $\alpha_i \in \{0, 1\}$ and the task state;
$h \in \mathcal{H} = \{\mathrm{Fuel}, \mathrm{Burning}, \mathrm{Burnt}\}^G$ is the fire; $\rho \in \mathcal{P} = \mathbb{R}^G_{\geq 0}$ is the smoke field; $c \in \mathcal{C} = \{\mathrm{intact}, \mathrm{collapsed}\}^G$ is the structure; $k \in \mathcal{K}$ is the realized communication graph on I; and the last factor is the step index t, with horizon T. The transition kernel factorizes as

$$\begin{aligned} T_\theta(s' \mid s, a) = \underbrace{T_C(c' \mid c, x)}_{\text{collapse}} \underbrace{T_H^{\beta,\kappa_A}(h' \mid h, c, c')}_{\text{fire; Coupling A}} \underbrace{T_\rho(\rho' \mid \rho, h')}_{\text{smoke}} \\ \times \underbrace{T_X(x' \mid x, a, h', c')}_{\text{agents}} \underbrace{T_K^{\delta}(k' \mid x')}_{\text{comms}}, \end{aligned}$$

and agent i's observation is drawn from the post-step state, $o'_i \sim O^i_{\kappa_B}(\cdot \mid x', h', \rho', c', k')$. The dependence of T_H on the collapse increment $c' - c$ is Coupling A (§5); the dependence of $O^i_{\kappa_B}$ on the fire, through the smoke field ρ' it produces, is Coupling B (§6). A training mixture (§9.1) draws θ once per episode, at reset. The slopes γ_p and γ_n (§9.2, Appendix H) are unrelated to this discount.

Proposition 1 (Well-posedness and Markov closure). For every θ, the process $s_t = (x_t, h_t, \rho_t, c_t, k_t, t)$ under any joint policy is a Markov chain, and $\mathcal{M}_\theta$ is a well-defined Dec-POMDP: finite-horizon optimal joint policies exist, and the standard value definitions apply.

The proof (Appendix A) composes the five kernels in order. Because ρ is real-valued the state space is not finite, but the states and observations reachable within the horizon form finite sets, and restricted to them $\mathcal{M}_\theta$ is a finite Dec-POMDP.

Remark 1 (Hardness). For two or more agents, the finite-horizon Dec-POMDP decision problem is NEXP-hard, and NEXP-complete when the horizon is below the number of states (Bernstein et al., 2002)[37]. This bounds the worst case over the class and says nothing about a particular instance; it licenses the expectation that no general exact planner will scale, which is why we learn policies rather than plan.

Listing 1, the body of `step`, follows the factorization; the link graph is drawn on its own stream with no hazard argument, and the reward line reads only the task reward and the two death counts. In the listing, `tl` is the episode's θ, and no configuration here uses the early `frozen` control branch. The observation is then drawn from the new state on its own stream (`che/env/env.py:394-396`).

Listing 1. One environment step in the order of Proposition 1.

```
# che/env/env.py:298-358 (verbatim)
    k_struct, k_seed, k_fire = jax.random.split(key, 3)

    # 1. c' ~ T_C(c, x): reads *pre-step* occupancy.
    occ_pre = occupancy_grid(state.agent_pos,
state.agent_alive, cfg.grid_size)
    structure_new = structure_step(
        k_struct,
        state.structure,
        state.weak,
        occ_pre.astype(jnp.float32),
        lambda_0=th.lambda_0,
        lambda_load=th.lambda_load,
    )
    collapse_increment = (structure_new == COLLAPSED) &
(state.structure == INTACT)

    # 2. h' ~ T_H(h, c, c'): CA spread, then the Coupling A
impulse from the
    # collapse increment (seeded cells are Burning in h',
spread next step).
    seed_mask = coupling_a_seed_mask(
        k_seed, collapse_increment, kappa_A=tl.kappa_A,
r_seed=th.r_seed
    )
    hazard_ca = hazard_step(k_fire, state.hazard,
beta=tl.beta, iota=th.iota)
    if cfg.hazard_mode == "frozen":
        # M1.3: h is frozen — the CA/seed draws above still
happen and are
        # discarded, so dynamic<->frozen with the same key
share every
        # other stream bitwise (invariant #3). Frozen Burning
cells stay
        # Burning: lethal and smoke-emitting, but never
spread or burn out.
        hazard_new = state.hazard
        seeded_ignitions = jnp.zeros_like(seed_mask)
    else:
        hazard_new = seed_ignitions(hazard_ca, seed_mask)
        seeded_ignitions = hazard_ca != hazard_new  # Fuel
cells Coupling A lit

    # 3. rho' from h' (Def. 6; smoke outlives flame).
    smoke_new = smoke_step(state.smoke, hazard_new,
sigma_s=th.sigma_s, eta=th.eta)

    # 4. x' ~ T_X + task dynamics. M1.1: lethality/blocking
against h'/c'.
    # DECISION: an agent disabled this step does not collect
food this step
    # (it died before/on arrival) — occupancy filters on the
post-death
    # alive vector, keeping "dead agents never collect"
exact.
    pos_new, alive_new, deaths_fire, deaths_collapse,
blocked_moves = agent_step(
        state.agent_pos,
        state.agent_alive,
        actions,
        hazard_new,
        structure_new,
        collapse_increment,
        cfg.grid_size,
    )
    occ_post = occupancy_grid(pos_new, alive_new,
cfg.grid_size)
    food_new, task_reward = task_step(state.food, occ_post)
    # Def.-2-compliant death penalty: reads only the alpha
transition (an X
    # variable) — never hazard/smoke/structure directly.
    reward = task_reward - th.death_penalty * (deaths_fire +
deaths_collapse).astype(
        jnp.float32
    )

    # 5. k' ~ T_K(x'): comms channel (Def. 7, M5.0). Reads
post-step
    # positions/aliveness only — never h', rho' or c' — and
draws from its
    # own fold_in stream, so delta cannot perturb any kernel
above.
    links, links_in_range = _comms_obs(
        jax.random.fold_in(key, _COMMS_STREAM), pos_new,
alive_new, cfg, tl.delta
    )
```

3.2 The fire and its smoke

Definition 3 (Fire kernel). Each cell is Fuel, Burning or Burnt. At each step, independently for each ordered pair of a Burning cell g and a Fuel von Neumann neighbour $g' \in N(g)$, the cell g' ignites with probability β. A Burning cell becomes Burnt after exactly one step, and Burnt is absorbing. Fuel cells may also ignite spontaneously at rate $\iota \geq 0$ per step; CHE uses $\iota = 0$.

Every episode starts from an all-Fuel grid with one Burning cell at a uniformly random position (`che/env/env.py:241-243`). One uniform per cell and incoming direction realizes the per-edge Bernoulli variables of Proposition 2's coupling exactly (`che/env/hazard.py:8-16`; Appendix A), and cells outside the arena never burn.

Proposition 2 (Bond percolation). Under Definition 3 with $\iota = 0$ and a single ignition at the origin, the set of eventually-Burnt cells is equal in distribution to the open cluster of the origin in i.i.d. bond percolation on (G, E) with edge-open probability β.

Here E is the arena's set of von Neumann edges. The proof (Appendix A) is a short coupling, the correspondence between an epidemic with a constant infectious period and bond percolation described by Grassberger (1983)[9]. Its hypotheses hold for CHE's fire when Coupling A is off, since collapses otherwise add ignitions.

Smoke outlives flame. Per cell,

$$\rho' = e^{-\eta}\rho + \sigma_s \mathbf{1}[h' = \text{Burning}],$$

with emission $\sigma_s = 1.0$ and decay $\eta = 0.5$, held fixed by convention and never calibrated. A cell emits σ_s once, in the step it burns, and its smoke then decays geometrically, with no spatial smoothing (`che/env/hazard.py:67-79`).

3.3 Agents, deaths and a hazard-blind reward

The agents kernel $T_X(x' \mid x, a, h', c')$ acts in three deterministic stages that consume no random numbers (`che/env/env.py:159-173`). An alive agent on a cell that collapses in this step is disabled before it moves. Survivors take one of 5 actions, stay or a von Neumann move clipped to the grid, and a move into a collapsed cell is cancelled. An agent whose new cell is Burning in h' is then disabled, so a cell igniting under a standing agent kills it; Burning cells can be entered, and avoidance must be learned. A disabled agent stays disabled, holds still, collects nothing and disappears from the others' observations.

Definition 2 (Ambient hazard). The hazard is *ambient* if and only if (i) and (ii) hold.

(i) *Reward independence:* the reward is a function of task variables and of the agents' alive transition only,

$$R(s, a, s') = R_{\text{task}}(\text{task}(x), a, \text{task}(x')) - d_p\,|\{i : \alpha_i = 1,\ \alpha$$

where $\text{task}(x)$ denotes the task variables (in CHE, the food grid and the alive agents' positions) and $d_p \geq 0$ is a fixed death penalty. No term of R references h, ρ or c, and the hazard appears in no auxiliary cost or constraint channel: no CMDP constraint, no Lagrangian penalty, no safety critic.

(ii) *Non-adversarial:* T_H is a fixed stochastic kernel; nothing in the environment optimizes, learns or best-responds.

Under clause (i), survival is learned through two hazard-blind routes: a dead agent collects no further task reward, and each death costs a fixed $d_p = 0.5$, whatever its cause, in every configuration reported here (the code's default is 0). Fire and collapse reach return only through the transition and observation kernels. The penalty prices an outcome, not the hazard. A hazard-priced reward signals the hazard's state whether or not anything is harmed: JaxWildfire, for example, penalizes its agent at every step in proportion to the number of burning cells (Çakır et al., 2025)[13]. At $d_p = 0$ only the first route would remain.

Listing 2's signature admits only the food and occupancy grids, so no fire, smoke or structure argument can reach the task reward.

Listing 2. The task reward reads food and occupancy only.

```python
# che/env/tasks.py:31-43 (verbatim)
def task_step(food: jax.Array, occupancy: jax.Array) ->
tuple[jax.Array, jax.Array]:
    """Collection: an occupied food cell yields +1 team
reward and empties.

    Multiple agents on one food cell still collect a single
item (+1).
    Returns (food', team_reward). Reads task variables and
agent occupancy
    only — no hazard/smoke/structure arguments, by design
(Def. 2).
    """
    chex.assert_type(food, jnp.uint8)
    chex.assert_equal_shape([food, occupancy])
    collected = (food == 1) & occupancy
    reward = collected.sum().astype(jnp.float32)
    food_new = jnp.where(collected, jnp.uint8(0),
food).astype(jnp.uint8)
    return food_new, reward
```

Two tests check clause (i). One asserts identical rewards, with and without the death penalty, for states that differ only in fire, smoke or structure, every difference far from all agents so that the alive transition is pinned equal (`che/tests/test_reward_independence.py:6-15`). It shows only that the reward reads no hazard state directly, since the fire is meant to change return through deaths. The other kills a standing agent by fire and, separately, by collapse, with identical task state, and asserts a reward of $-d_p$ both times (`che/tests/test_lethality.py:143-162`).

Clause (i) also fixes how §9's two co-primary outcomes enter learning: survival is a penalized channel, each death costing d_p, and completion a rewarded one, each collected item earning +1. §9.2 reads any asymmetry between them as one between a penalized channel and a rewarded one.

The one ablation of d_p is from the calibration phase, with this scope: the fire alone with no coupling on, 500 updates, 3 seeds per cell, an observation encoding that predates the one used here, and means over the final training updates. Raising d_p from 0 to 0.5 moved High survival from 0.575 to 0.866 and High completion from 0.765 to 0.821, moved Medium survival from 0.931 to 0.951 with completion 0.750 at both, and left Low tied, 0.991 against 0.992. It was not re-run under the current observation or with the couplings.

Clause (ii) does not make the hazard independent of the agents: a weak cell with an agent on it is nine times as likely to collapse in a step as an empty one, and a collapse can seed fire (§5.1).

3.4 Nested models: ablations are exact by construction

Each stressor is a parameter of one transition or observation kernel, and setting it to zero recovers a nested special case of the same

model.

stressor	parameter	lives in	off ($= 0$) recovers	asserted by
Coupling A (Definition 5, §5)	κ_A	T_H, through $c' - c$	a fire blind to the structure	`che/test`
Coupling B (Definition 6, §6)	κ_B	O^i	the observation before masking, plus an all-ones visibility plane	Listing 3
communication denial (Definition 7, §7)	δ	T_K	free, range-limited communication	`che/test`

The fire's β is a parameter of the same kind, positive in every configuration and set by severity (§4).

One engineering rule makes the nesting exact: every stochastic branch draws its uniforms unconditionally and compares them against a probability that may be zero. The draws for Coupling B's reveal, the links and the per-episode mixture use streams derived from the reset or step key by a fixed tag, so adding them left the existing streams untouched. Setting a stressor to zero therefore changes comparisons only, and the same keys reproduce the nested model's trajectory bit for bit.

Listing 3 runs the environment at $\kappa_B = 0$ with and without masking, under the same keys, and asserts bitwise-identical fire, smoke, structure, food, positions, alive flags and rewards. The rest of the test (`che/tests/test_coupling_b.py:143-156`) steps both by hand and asserts identical content planes, an all-ones visibility plane and a zero masked share.

Listing 3. With $\kappa_B = 0$, trajectories and rewards are bitwise those of the environment before masking.

```
# che/tests/test_coupling_b.py:131-142 (verbatim)
def test_kappa_b_zero_bitwise_recovers_obs_v2():
    """Nesting (M4.1): obs v3 at kappa_B = 0 is the pre-
masking env plus a
    constant all-ones visibility plane — state trajectories,
rewards, and
    content planes bitwise identical to obs v2 under the same
keys."""
    theta = dataclasses.replace(_HOT, kappa_B=0.0)
    cfg3 = EnvConfig(grid_size=L, n_agents=4, horizon=64,
theta=theta)
    cfg2 = dataclasses.replace(cfg3, obs_version=2)
    a = _traj(theta, cfg=cfg3)
    b = _traj(theta, cfg=cfg2)
    _assert_bitwise_equal(
        a, b, ("hazard", "smoke", "structure", "food", "pos",
"alive", "reward")
    )
```

With $\kappa_A = 0$, collapse activity leaves the fire and smoke bitwise unchanged (`che/tests/test_nesting.py:78-88`); at $\delta = 0$, the environment reproduces, on the CPU, a trajectory digest recorded before the communication channel existed (`che/tests/test_comms.py:272-319`). A structural test also asserts that the traced computation keeps its random-number primitives at parameter value zero, catching a gate such as `if kappa_A > 0:` that behavioural tests miss (`che/tests/test_nesting.py:222-240`).

3.5 The instance, training and metrics

Every trained policy and evaluation in this paper uses one instance:

a 64 × 64 grid, 12 agents, a horizon of 256 steps, a 9 × 9 egocentric crop and 32 food items. Food sits on distinct cells from reset and does not respawn, and an alive agent on a food cell collects it for a team reward of +1. The observation's 8 channels are six indicators (burning, burnt, food, weak, collapsed, alive occupancy), the raw smoke density and Coupling B's visibility plane, which marks the revealed cells. An own-state vector gives the agent's row and column over L, its alive flag and t over the horizon.

All agents share one actor-critic network over the crop, the own-state vector and the aggregate of delivered messages. It has two convolutions with 16 and 32 channels, a dense layer of 128 units, and separate actor and critic heads with a 64-unit layer each. A message head receives no gradient, so what it emits is a frozen-at-initialization projection of trained trunk features (§7).

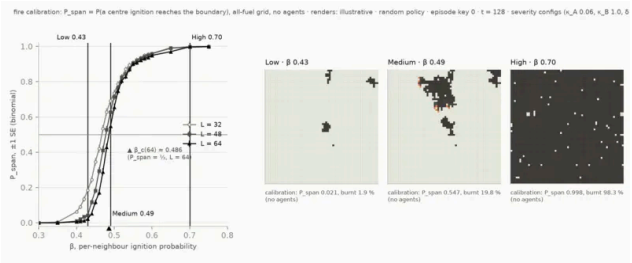
Training is the independent PPO of de Witt et al. (2020) [11], in the PureJaxRL style of Lu et al. (2022) [38], on the environment conventions of JaxMARL (Rutherford et al., 2024) [16]. Each seed trains one shared-parameter policy, and each agent's transitions are independent samples under the team reward. Each update collects 128 steps from 256 parallel environments, 32,768 environment steps, each advancing all 12 agents. ISO, ISO-4 and JOINT (§9) each train for T* = 1000 updates, or 32,768,000 environment steps, with the hyperparameters of Appendix B.

The co-primary outcomes of §9 are per-episode. *Completion* is $1 - F/32$, where F is the number of food items left at the episode's last step; *survival rate* is the fraction of the 12 agents alive at the end. Episodes always run to the horizon, even when every agent has died (`che/env/env.py:361`; `che/eval/harness.py:44-45`). Deaths are also recorded by cause, as diagnostics.

4 Severity calibration

Severity in CHE is placed by a measured dynamical phase of the fire: the three levels sit below, near and above the implemented kernel's measured transition (Figure 5).

Figure 5. Severity as a measured phase. Left: the spanning probability, the chance that one centre ignition reaches the arena boundary, against β on an all-fuel grid with no agents, for three arena sides (512 seeds per point). Vertical rules mark the locked levels; the triangle marks the largest arena's pseudo-critical point. Right: one random-policy episode per level under the severity configurations, with δ 0; only the fire and its smoke (hatching) are drawn. Neither the R_L curves nor the locked critical point β_c is shown (§4.3). Status: calibration, no agents (curves) · illustrative, one episode each (renders) · Interact: play or drag the time slider; click a panel.



Static companion of Figure 5 for the printed PDF.

4.1 Phase structure of the idealized kernel

Corollary 1 (Phase structure). On $\mathbb{Z}^2$ with the von Neumann neighbourhood, the fire of Definition 3 with $\iota = 0$ and a single ignition at the origin inherits the phase structure of bond percolation (Proposition 2, whose coupling holds on any graph):

1. Its critical point is $\beta_c = 1/2$ exactly, the critical probability of bond percolation on the square lattice (Kesten, 1980) [10].

2. For $\beta > \beta_c$ the fire spreads indefinitely with positive probability. On that event the cells ignited by step n are those within n open edges of the origin. Scaled by $1/n$, they converge almost surely in Hausdorff distance to a deterministic convex compact set with non-empty interior, invariant under permutations and reflections of the coordinates (Garet & Marchand, 2004; their Corollary 5.4) [39]. The growth is therefore linear in n, at a speed given in each direction by the extent of that set.

Both items concern the idealized kernel (von Neumann neighbourhood, constant burn time, homogeneous fuel, no wind) on the infinite lattice. CHE's fire is that kernel on a finite arena, with ignitions added by Coupling A (§3.2), so the levels are placed against a measured finite-size threshold and, at High, a measured front speed (§4.3).

4.2 Severity as phase

Definition 4 (Severity regimes). Fix the arena side L. Severity levels are defined by the position of the implemented kernel relative to its measured critical point $\hat{\beta}_c$, through the correlation length $\xi(\beta)$, the typical extent of a fire cluster:

- *Low*, subcritical, $\xi(\beta) \ll L$: fire clusters are local and short-lived, and survival is a local-avoidance problem.
- *Medium*, near-critical, $\xi(\beta) \sim L$: cluster sizes are scale-free up to the arena's size, fluctuations are maximal, and a single ignition can, with non-negligible probability, cascade to arena scale; survival is a global-anticipation problem.
- *High*, supercritical: fire fronts move at a speed comparable to the agents', and survival is a race.

The correlation length motivates the levels; operationally, each level is the observable band it was locked to (§4.3). As qualitatively different phases of one kernel, the levels make training at Low and High and evaluating at Medium (§9.1) a test of transfer to a phase neither arm trained in, though in the scalar β it remains an interpolation (§10). Definition 4 also carried an expectation, on record before its test (§4.4), that per-episode outcome variance would peak at Medium.

4.3 Measuring the threshold and locking the levels

The calibration ran the fire alone on an all-Fuel grid at $L \in \{32, 48, 64\}$, with 512 seeds per (L, β). A span sweep over 35 values of β (0.05 to 0.95 coarsely, 0.40 to 0.60 in steps of 0.01), each from one centre ignition for $T_{\max} = 4L$ steps, measured P_{span}, the burnt fraction and the front speed. A crossing sweep ignited the full left column and recorded the left–right crossing probability $R_L(\beta)$.

The severity lock fixed $\hat{\beta}_c = 0.500 \pm 0.005$, from logistic fits to R_L with centres 0.4985, 0.5006 and 0.4999 at $L = 32, 48, 64$, and a self-duality check $R_L(0.500) = 0.488, 0.502, 0.494$. The lock record states that they were recomputed independently from the committed raw data; no committed script or artifact produces the logistic centres. Among the committed estimators, pairwise crossings of R_L average 0.504; the family spans [0.480, 0.513], with consensus mean 0.499. All of them bracket Kesten's ½. That agreement validates the implementation; it is not a finding about fire, since Proposition 2 and Corollary 1 predict it for this kernel.

Each level was then locked to an observable band, with its measured values ($L = 64$, 512 seeds, no agents):

level	β	phase	P_{span}	burnt fraction	band the level was locked to
Low	0.43	subcritical	0.021	1.9 %	$P_{\text{span}} < 0.05$ and burnt fraction in [1, 5] %
Medium	0.49	near-critical	0.547	19.8 %	P_{span} in [0.3, 0.7]
High	0.70	supercritical	0.998	98.3 %	front speed in [0.5, 1.0] cells per step, measured 0.83

At High the front moves at 83 % of the agents' speed of one cell per step, so it can be outrun, barely. Medium sits between the finite-size pseudo-critical point $\beta_c(64) \approx 0.486$, where $P_{\text{span}} = 1/2$ at $L = 64$, and $\hat{\beta}_c$. It is the severity of the held-out evaluation configuration of §9, $\theta^* = (\beta, \kappa_A, \kappa_B, \delta)$ = (0.49, 0.06, 1.0, 1.0); the compositional test trains at Low and High only.

4.4 Re-testing the Medium prediction

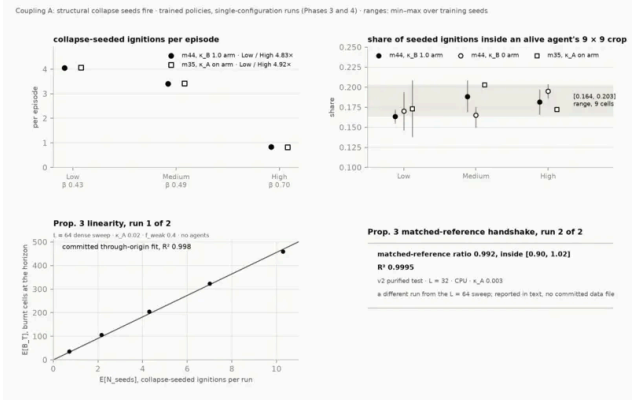
The calibration-phase probe at $d_p = 0.5$ (the fire alone, 500 updates, three seeds per severity) had saved no checkpoints. The re-test at fixed policy re-ran it from the same configurations and seeds (equivalent, not bitwise identical) and evaluated each final checkpoint on 512 episodes. For survival the prediction is refuted: per-episode variance at fixed policy is monotone in severity, 0.00091, 0.00709 and 0.00885 at Low, Medium and High. For completion it is not confirmed: Medium and Low tie within bootstrap intervals, 0.01245 against 0.01235, both above High. The environment-level mechanism is confirmed: with no agents, the burnt fraction's variance peaks near $\beta \approx 0.53$, just above Medium, and ranks Medium above High above Low at the locked levels.

The re-test report offers an account it did not test: outcome variance is roughly the environment's variance, which peaks near criticality, times the policy's sensitivity to it, which grows with severity. The refutation is scoped to survival, for these fire-only policies at this budget; §10 states what siting the held-out test at Medium costs.

5 Coupling A: collapse seeds the fire

Coupling A lets structural collapse feed the fire. Figure 1's event track marks one random-policy episode's collapses and the ignitions they seeded (press `]` in Figure 1 to step between them); Figure 6 counts them under trained policies.

Figure 6. Coupling A under trained policies, and the two Proposition 3 runs. *Top:* collapse-seeded ignitions per episode by severity, and their share inside an alive agent's crop, in the Coupling-B and Coupling-A calibration grids (labelled m44 and m35; Phases 3 and 4 are calibration stages). Whiskers are min–max over training seeds; neither grid has identical reruns, so nothing shown has a floor. *Bottom:* the two runs, kept apart, neither at the locked parameters; the second (v2, a revised test in the suite) has no committed data file, so its panel gives values. Status: trained policies, 500 updates; seed ranges · Interact: hover or tab to a point; open the table view.



Static companion of Figure 6 for the printed PDF.

5.1 Definition and implementation

Definition 5 (Coupling A, as implemented). Let $w(g) \in \{0, 1\}$ mark weak terrain, a spatially clustered fraction f_{weak} of the cells drawn once per episode, and let $o_t(g) \in \{0, 1\}$ indicate that at least one alive agent stands on cell g before the step. An intact cell collapses at step t with probability

$$\lambda_t(g) = \big(\lambda_0 + \lambda_{\text{load}}\, o_t(g)\big)\, w(g),$$

so a cell that is not weak never collapses, and collapse is absorbing. Coupling A then ignites, with probability κ_A, each cell that is still Fuel after this step's spread and lies within Chebyshev distance r_A of a cell that collapsed in this step. The fire kernel thus reads only the collapse increment $c' - c$, and a seeded cell is Burning in h'. Setting $\kappa_A = 0$ recovers the uncoupled fire kernel $T_H^{\beta,0}$, so the ablation "no Coupling A" is a nested model of the full system.

The theory states the impulse per collapse event; the implementation draws one uniform per cell, and the two rules coincide wherever seeding neighbourhoods do not overlap. The values, locked on 2026-07-21, are $(f_{\text{weak}}, \lambda_0, \lambda_{\text{load}}, \kappa_A, r_A) = (0.15,\ 5 \times 10^{-5},\ 4 \times 10^{-4},\ 0.06,\ 1)$, so the seeding neighbourhood is a 3 × 3 block.

Agents influence collapse, weakly and indirectly. Occupancy enters as a yes-or-no load, so a weak cell with any agent on it collapses nine times as often as an empty one. Without load, the locked parameters give about 7.81 collapses per episode. Load adds 0.4 to 1.4 % to that count under trained policies (the Coupling-A calibration grid of §5.3) and 7.4 to 8.3 % under random ones, which is 6.9 to 7.6 % of all collapses. Igniting a cell is not among the agents' actions, but the fire is not independent of them.

Listing 4. Coupling A's seeding mask, with its uniforms drawn whatever κ_A is.

```python
# che/env/structure.py:83-102 (verbatim)
def coupling_a_seed_mask(
    key: jax.Array,
    collapse_increment: jax.Array,
    *,
    kappa_A: float,
    r_seed: int,
) -> jax.Array:
    """Coupling A (Def. 5): each cell in the seeding
neighborhood N_A of a
    new collapse (Chebyshev radius r_seed) ignites
independently w.p.
    kappa_A. `collapse_increment` is the boolean mask of c' -
c, i.e. cells
    that collapsed *this step* — T_H reads the increment, not
the stock.
    Returns the candidate ignition mask;
`hazard.seed_ignitions` applies it
    to Fuel cells only.
    """
    chex.assert_rank(collapse_increment, 2)
    chex.assert_type(collapse_increment, jnp.bool_)
    # Invariant #3: uniforms drawn even when kappa_A == 0.
    u = jax.random.uniform(key, collapse_increment.shape)
    near_new_collapse = dilate(collapse_increment, r_seed)
    return near_new_collapse & (u < kappa_A)
```

The uniforms `u` are drawn before anything reads κ_A, so $\kappa_A = 0$ is the nested model bit for bit under the rule of §3.4. The mask then goes to `hazard.seed_ignitions`, which lights only cells that are still Fuel; this is where the fuel limitation of §5.3 enters.

5.2 The handshake with Proposition 3

Proposition 3 (Collapse events dominate hazard exposure). Work in the subcritical regime $\beta < \beta_c$ with $\iota = 0$, and suppose collapse events arrive at rate λ_A per step, each seeding one ignition at a uniformly located Fuel cell, with seeded clusters pairwise disjoint (the sparse regime, $\lambda_A T \chi(\beta) \ll |G|$, the number of cells). Then the expected burnt area B_T by time T satisfies

$$\mathbb{E}[B_T] = \lambda_A T\, \chi(\beta)\, \big(1 + o(1)\big), \qquad \text{and in general} \quad \mathbb{E}[B_T]$$

where $\chi(\beta)$ is the mean percolation cluster size. Consequently the marginal hazard exposure per unit of structural instability is $\partial\, \mathbb{E}[B_T] / \partial \lambda_A = T \chi(\beta)$.

The proof (Appendix A) grows each seed into a bond-percolation cluster by Proposition 2 and sums expectations; overlaps only remove double counting. With no primary ignition, as in both runs below, collapse is then the fire's only birth channel. In a finite protocol the equality is approached only with an unconditional cluster-mass reference, matched seed locations and ages, and sparse seeding; each correction has a known sign (Appendix A).

We tested the proposition in two separate runs, neither at the locked parameters, both without agents or primary ignition at the subcritical Low level ($\beta = 0.43$, $\iota = 0$). The first, a dense sweep on a 64 × 64 grid ($\kappa_A = 0.02$, $f_{\text{weak}} = 0.4$), fits burnt area against realized seeds through the origin: $R^2 = 0.998$ over a 14× range of the expected seed count. The second, a slower test in the suite on a 32 × 32 grid ($\kappa_A = 0.003$), divides its slope by a protocol-matched reference computed inside the test: 0.992, inside the band [0.90, 1.02], with $R^2 = 0.9995$.

5.3 Under trained policies the coupling limits itself

In the Coupling-B calibration grid (500 updates at each severity, κ_B off and on, Coupling A on), the $\kappa_B = 1.0$ arm records 4.05, 3.41 and 0.84 collapse-seeded ignitions per episode at Low, Medium and

High, a 4.83-fold fall. The Coupling-A calibration grid (κ_A off and on, with other policies, trained on the observation before masking) replicates it: 4.08, 3.42 and 0.83, a 4.92-fold fall.
Collapse never reads β, so what changes with severity is where the seeds land: at High the fire burns 98.3 % of the arena in calibration, and the Fuel-only rule discards most seeding attempts. This concerns ignitions only: the Coupling-A grid's survival contrasts, on two seeds per arm and with no floor, cannot locate the coupling's largest effect on outcomes.

This fuel limitation was on record before any trained policy was measured. At the Coupling-A lock, the random-policy calibration put seeded ignitions at High about 5.7 times below Low at every κ_A tried, and the lock recorded it as the fuel-exhaustion self-limitation of the ignition channel. A later prediction was written after the trained results existed and before its own analysis. It held that Coupling A is fuel-limited at High, that the co-active count of §6.5 need not be monotone in severity, and that Medium may be its peak. The 4.83-fold fall confirms the first, which earlier reports had already answered. The count is monotone non-increasing and Low and Medium are indistinguishable, so the second is not supported and the third is refuted.

The fall happens upstream of the agents. Across all nine measured cells (two arms of the Coupling-B grid and one of the Coupling-A grid, at three severities), the share of seeded ignitions inside an alive agent's crop lies in [0.164, 0.203].

At Low the coupling moves survival in the Coupling-A grid off its ceiling, from 0.979 to 0.961, as fire deaths per episode rise from 0.062 to 0.229. Weak-cell occupancy is higher by 0.054 with the coupling on, so the policies do not avoid weak cells, which two seeds per arm only suggest. No contrast in this section has a floor on its own artifact, since neither grid has identical reruns; any floor beside them is a reference scale (cross-artifact); see §8.6.

6 Coupling B: smoke attenuates perception

Coupling B makes what an agent perceives depend on the hazard's own state. Figure 1 opens on agent 7 at t = 128 and shows that agent's observation as produced, the true state of the same 9 × 9 cells, and the probability that each cell was seen.

6.1 Definition and implementation

Definition 6 (Coupling B: Beer–Lambert observation, as implemented). Smoke is the per-cell state field ρ' of §3.2, updated after the fire with no spatial smoothing, so it outlives flame. For agent i at x_i and a cell y of its crop, let $\bar{\rho}_i(y)$ be the mean of ρ' at the four points $x_i + t_s(y - x_i)$, $t_s = (s + \frac{1}{2})/4$ for $s = 0, \ldots, 3$, each read at the nearest cell (smoke outside the grid reads 0). The transmittance is

$$\tau_i(y) = \exp\big(-\kappa_B \, \|y - x_i\|_2 \, \bar{\rho}_i(y)\big),$$

a midpoint-rule form of $\exp(-\kappa_B \int_{x_i \to y} \rho \, d\ell)$ along the line of sight. Each crop cell is revealed independently with probability $\tau_i(y)$: a revealed cell passes its seven content planes, an unrevealed one passes zeros, and an eighth plane records the realized reveal, so the agent can tell "unseen" from "empty". The own cell has distance 0 and is always seen. Setting $\kappa_B = 0$ gives $\tau \equiv 1$ and recovers hazard-independent observation.

Because ρ' is hazard state, perception degrades where and when the fire is active; Beer–Lambert attenuation is the standard optical model for light through smoke.

Listing 5. Coupling B in the observation kernel: transmittance, the reveal draw and the visibility plane.

```
# che/env/observation.py:300-310 (verbatim)
        tau = transmittance(
            state.smoke, state.agent_pos,
kappa_B=state.theta_live.kappa_B, k=k
        )
        reveal = jax.random.uniform(key, tau.shape) < tau  #
P(seen) = tau
        grid = jnp.concatenate(
            [
                grid * reveal[..., None].astype(jnp.float32),
                reveal[..., None].astype(jnp.float32),
            ],
            axis=-1,
        )
```

The `reveal` line draws its uniforms whatever κ_B is, so at $\kappa_B = 0$ the mask is all ones and the environment is bitwise the one before masking (§3.4, Listing 3). The draw has its own random stream, and κ_B is read per episode from the state, which is how a training mixture varies it.

6.2 What attenuation does to the value of adaptivity

Theorem 1 (Memorization gap, and its erosion by Coupling B). Consider the two-corridor environment $E_{2C}(\kappa_B)$: from a start cell, a path of d steps leads to a branch b, from which two disjoint corridors L and R of length ℓ lead to the goal; the horizon is $T = d + \ell$ (zero slack: a wrong commitment cannot be undone). At $t = 0$ nature draws $Z \in \{L, R\}$ uniformly; corridor Z burns at depth $\ell_f \geq 1$ beyond b, and an agent entering it is disabled. The reward is 1 for reaching the goal by T, else 0. At each step $t \leq d$ the agent receives the signal $Y_t = Z$ with probability $e^{-\kappa_B(d-t+\ell_f)}$ (Beer–Lambert over the current distance to the fire, unit smoke density), else $Y_t = \varnothing$; signals are independent across time and remembered. Let

$$q(\kappa_B) = 1 - \prod_{j=\ell_f}^{d+\ell_f} \left(1 - e^{-\kappa_B j}\right)$$

be the probability of being informed by the commitment point. Then (1) the optimal value is $J^*(\kappa_B) = \frac{1}{2} + \frac{1}{2}q(\kappa_B)$; (2) the policies optimal for the fixed-map variant ($Z \equiv z_0$ constant) include the signal-blind policy "always take the corridor other than z_0", whose value on E_{2C} is $\frac{1}{2}$ for every κ_B, so the memorization gap is $J^*(\kappa_B) - J^{\text{static}} = \frac{1}{2}q(\kappa_B) > 0$ for all finite κ_B; and (3) q is continuous and strictly decreasing in κ_B, with $q(0^+) \to 1$ and $q(\kappa_B) \to 0$ as $\kappa_B \to \infty$, so under total perceptual denial the optimal adaptive policy is no better than the memorizing one.

In one toy (proof in Appendix A), a memorizing policy loses exactly half the probability of being informed, and the hazard's own attenuation shrinks that loss continuously to zero; the theorem says nothing about what a learner does in CHE.

The theorem also tests the code. The toy and the observation kernel share one transmittance function, and the toy's information probability agrees at every grid point across Monte Carlo, the closed-form product and the full observation pipeline. The measured optimal value matches $\frac{1}{2} + \frac{1}{2}q$ under a three-condition gate: largest $|z|$ 2.11, $\sum z^2 = 6.55$ on 8 degrees of freedom ($p = 0.586$), and mean z −0.44. The memorizing policy stays flat at ½, $J^*(0) = 1.0000$, and $J^* - \frac{1}{2} = 0.0000$ at $\kappa_B = 8$, where $q = 0.0056$.

6.3 The lock: three bands that did not intersect

The strength κ_B was to be locked where three bands set before calibration intersect. They were a masked share of the crop at Medium in [0.15, 0.45], a probability in [0.4, 0.7] that a Burning cell three cells away is revealed at Medium, and a toy information probability q in [0.3, 0.7]. They do not intersect. The masked-share band cannot be reached at Medium: the masked share can only approach the share of crop cells with smoke on their line of sight, 0.130 there under a random policy, below the band's lower edge of 0.150. The same κ_B attenuates the arena's extended smoke more strongly than the toy's single smoke source, so the detection and toy bands are disjoint by a factor of 1.35 under a random policy and 1.15 under probe policies. On 2026-07-27, $\kappa_B = 1.0$ was locked on the detection band alone; the masked-share band was retired, not widened, and the toy band kept as a consistency check.

6.4 The cost is paid in survival, at High

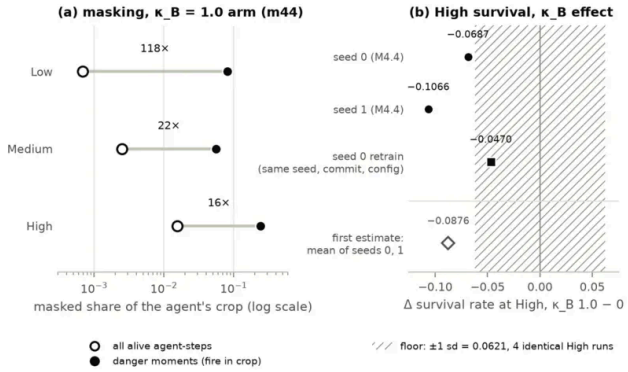


Figure 7. Coupling B: where the masking falls, and the High survival contrasts.

(a) Masked share of the crop in the coupled arm of the Coupling-B calibration grid (labelled m44 and M4.4), over all alive agent-steps and at danger moments. (b) The High survival contrasts between the arms: three replicates, then the first estimate. The hatched band, labelled floor, is a reference scale (cross-artifact). The figure cannot show intervals, completion or Medium. Status: trained policies, 500 updates · static.

At High, three replicate contrasts of survival between $\kappa_B = 1.0$ and $\kappa_B = 0$ are negative three times out of three (Figure 7(b); trained policies, 500 updates, $d_p = 0.5$, Coupling A on). They are −6.87 and −10.66 points for the grid's two seeds and −4.70 points for a same-seed retrain, each one run per cell and without an interval. The grid's first estimate, the mean of its two seeds, was −8.8 points. The High survival floor, sd 6.21 points, was measured later on four identical runs of a separate artifact, so it is a reference scale (cross-artifact). The first estimate is 1.41 times this floor and the replicates' mean about 1.19 times it. We report a consistent effect of modest size; the first analysis graded it "strong" against a two-seed spread, a label retracted under the rule of §8.1. Completion is not resolved, since its contrast flipped sign on the same-seed retrain.

At Medium, the severity of the held-out test configuration (β = 0.49), Coupling B's own survival effect in the same grid was −0.03 points. It was judged against a three-seed spread, $\sigma_{\text{seed}} = 0.72$ points, with no measured floor; the Medium survival floor measured later is 1.29 points. Coupling B's measured survival cost is therefore at High, while §9's test runs at Medium (§10, item 1).

The masking concentrates on moments of danger, alive agent-steps with a Burning cell inside the agent's crop. In the $\kappa_B = 1.0$ arm the masked share of the crop at danger moments is 0.0809, 0.0560 and 0.2424 at Low, Medium and High, against 0.00068, 0.00249 and 0.01559 over all alive agent-steps: 118, 22 and 16 times more concentrated (Figure 7(a)). Within trained policies at 500 updates the attenuation was not behaviourally suppressed: the coupled policies' masking ceiling, which does not depend on κ_B, sat at or above its random-policy value at every severity.

6.5 The co-active counter, and what it can see

The environment logs, at every step, the number of collapse-seeded ignitions inside an alive agent's crop, within Chebyshev radius 4 of its post-step position. It was meant to catch both couplings at once, but only Coupling A produces those ignitions and κ_B does not enter the count, so the counter is co-active only in all-on configurations. In a training mixture it reads the Coupling-A marginal alone, which is why the mediation analysis registered on it is void (§9.2).

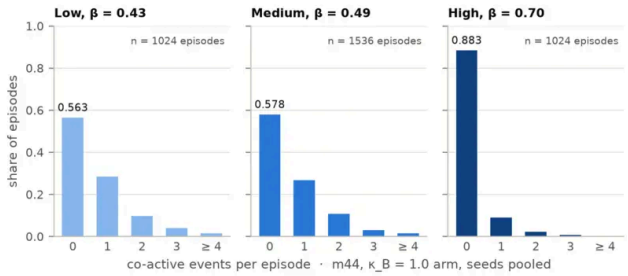


Figure 8. The co-active counter per episode, by severity.

Share of episodes with each number of co-active events in the coupled arm of the Coupling-B calibration grid (labelled m44), seeds pooled; both couplings are on in every cell. The figure cannot show a floor, since none exists for this counter, or a training mixture (§6.5). Status: descriptive; trained policies, 500 updates; no floor · static.

In these all-on cells the counter is rare and bursty (Figure 8). The share of episodes with no co-active event is 0.563, 0.578 and 0.883 at Low, Medium and High, the mean at Medium is 0.64 events per episode, and the variance-to-mean ratio, 1.31 to 1.40, exceeds the Poisson value of 1. Low and Medium are indistinguishable and High is lower, because seeded ignitions fall while the near-agent share stays in its range (§5.3).

7 Communication denial: inert under this encoder at this scale

The third stressor denies links between agents; in CHE it had no measurable effect (Figure 9).

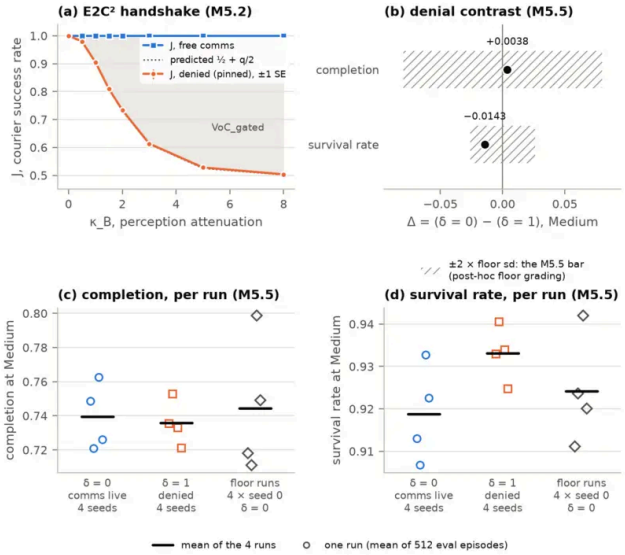


Figure 9. Communication in the toy, and denial in CHE at Medium.

(a) The courier variant of the corridor toy (labelled E2C² and M5.2), scripted policies: success with free and with denied communication on the no-idle schedule (pinned); the shaded gap is $\mathrm{VoC}_{\mathrm{gated}}$ (§7.2). (b–d) The denial study (M5.5): contrasts against bars (post-hoc floor grading), and the runs. No panel grades inertness or shows the content ablations, High severity, the VOID condition or a trained encoder. Status: descriptive · static.

7.1 The kernel

Definition 7 (Communication denial, as implemented). After the agents move, the comms kernel $T_K(k' \mid x')$ samples a directed link graph over the alive agents: for every ordered pair $i \neq j$ of alive agents,

$$\Pr\left[\, i \to j \text{ open} \,\right] = \mathbf{1}\left[\|x'_i - x'_j\|_\infty \leq R_{\mathrm{comm}}\right](1-\delta),$$

with $R_{\mathrm{comm}} = 16$ cells. Each agent emits a message $m_i \in [-1,1]^8$ at step t; at $t+1$ each receiver j gets the mean of the messages delivered over the links then realized, $\bar{m}_j = \sum_i k'_{ij}\, m_i \,/\, \max(1, \sum_i k'_{ij})$, which is the zero vector when j is isolated. T_K reads positions and aliveness only, so denial is mechanistically independent of the fire. One uniform per ordered pair is drawn at every step whatever δ is, so $\delta = 0$ gives the deterministic range graph and $\delta = 1$, the value locked on 2026-07-30, gives the empty graph, both bitwise. The theory's range-dependent link probability is implemented as this hard Chebyshev range.

Listing 6. Delivered messages carry no gradient.

```
# che/train/ippo.py:323-329 (verbatim)
        # Messages emitted this step are delivered next step.
Nothing crosses
        # an episode boundary (autoreset), and stop_gradient
states in code
        # what the batch-storage already implies: the channel
is not a
        # differentiable path (Q3 ruling; networks.py
docstring).
        messages = jax.lax.stop_gradient(
            jnp.where(done[:, None, None], 0.0, emitted)
        )
```

No gradient crosses the channel, so the message head, a tanh-bounded dense layer on the shared trunk, stays at its initialization for the whole run: an agent emits a frozen random projection of trunk features that are themselves trained. The receivers' weights on the message inputs are trained, so whether agents use messages can be tested; what to say was never learned.

7.2 When communication should matter

Remark 2' refines a result of Pynadath & Tambe [35]. Free communication lowers the complexity of the team decision problem when the world is collectively observable (their Theorem 4 and Table 2), and makes no difference when it is individually observable or not observable at all (their Theorems 5–6). Remark 2' concerns the value of communication under collective partial observability.

Remark 2'. Extend E_{2C} with a second agent. (i) Under a team-any reward, communication has zero marginal value when the agents are interchangeable, expendable (no death cost) and at least as numerous as the hypotheses: coverage then achieves $J = 1$ under total denial. When any qualifier fails, value returns; in general it scales with the deficit of agents relative to hypotheses, and a death cost, such as CHE's $d_p = 0.5$, prices the redundancy that substitution spends. (ii) In the courier variant, where only agent 1 can complete the task and it observes the other agent only through messages, the denied optimum within the no-idle policy class is $\frac{1}{2} + \frac{1}{2}q(\kappa_B)$; with slack ℓ_f, optimal denied play buys information by idling at the branch and achieves $\frac{1}{2} + \frac{1}{2}\tilde{q}(\kappa_B)$, where $\tilde{q}$ is the probability of being informed by the commitment point under the best idling schedule, with $\tilde{q} \geq q$ (equality iff $\ell_f = 0$), so

$$\mathrm{VoC}_{\mathrm{true}}(\kappa_B) = \tfrac{1}{2}\big(1 - \tilde{q}(\kappa_B)\big) \leq \tfrac{1}{2}\big(1 - q(\kappa_B)\big).$$

Both q and $\tilde{q}$ decrease in κ_B, so the value of communication increases with perception decay.

We ran the courier variant with scripted policies, delivering messages through the production link kernel and computing τ with the shared transmittance function, so the toy also tests both kernels. The free arm always succeeds, the no-idle denied arm matches $\frac{1}{2} + \frac{1}{2}q$, and the value of communication against that arm, $\mathrm{VoC}_{\mathrm{gated}} = \frac{1}{2}(1 - q)$, rises monotonically from 0 to 0.498 (Figure 9(a); Appendix A).

7.3 In CHE the channel carries nothing measurable

CHE's agents meet one of Remark 2'(i)'s qualifiers: they are twelve interchangeable agents sharing one set of parameters. Nothing establishes that they are at least as numerous as the hypotheses, and their deaths are priced ($d_p = 0.5$), so the clean-zero conditions are not fully met.

In a content ablation at Medium ($\delta = 0$, $R_{\mathrm{comm}} = 8$, two seeds, 500 updates), three arms of identical architecture pass live, zeroed or shuffled messages (the delivered aggregate set to zero with links kept; sender identities permuted within a step). No pair differs beyond a bar of twice the earlier Medium floor, about 3 points. The same ablation at High (three seeds, R_{comm} = 8 and 16) found that tripling the mean alive out-degree, from 1.01 to 2.99, changed nothing, ruling out "nobody was in range". The wider High floor puts that bar at about 11 points.

In the denial study (Medium, four seeds, $R_{\text{comm}} = 16$, message path live in both arms), $\delta = 0$ against $\delta = 1$ moves completion by +0.38 points against a bar of 7.99 points, and survival by −1.43 points against a bar of 2.60 points. Both bars are twice the floor sd measured on that GPU (Figure 9(b–d)). Every bar here is a post-hoc floor grading that a true effect of its size would clear only about half the time (§8.2), so none of these results is a point null. A further condition, on danger-moment outcomes, is VOID: its threshold, chosen at implementation, sat below the quantity's own run-to-run nondeterminism (§8.1).

The content ablations supply a mechanism: in the live arm a receiver can tell that it heard from someone without decoding anything; in the zeroed arm it cannot. That signal needs no encoder and went unused at both severities and, at High, at both ranges. The neglect is therefore on the demand side (the agents did not need the information) rather than the channel side (they could not read it), and the frozen encoder cannot account for it.

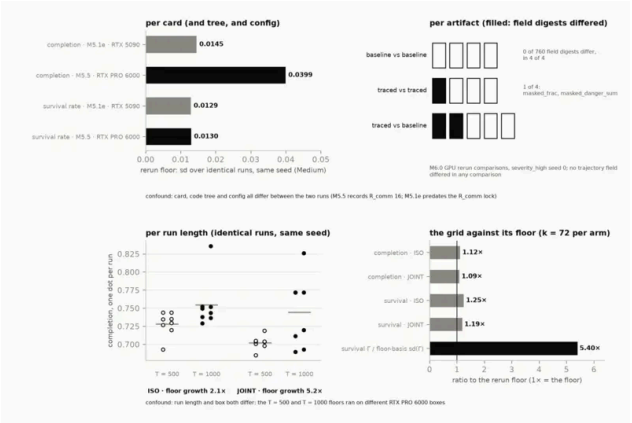
With deaths priced, the null also says that the value returning when a qualifier fails is weak at this scale and this death price. Gradient-shaped messaging is untested, and a bound measured on a frozen random encoder does not strictly bound a trained one. We claim only that communication denial is inert under a frozen random-projection encoder at this scale.

The composition §9 tests is thus, in effect, Coupling A × Coupling B, with $\delta = 1.0$ kept in θ^* for fidelity to the registration. The δ-only third of ISO's training is fire-only, and §9.1 gives the consequence.

8 Methodology: bars come with floors

GPU training is not reproducible run to run: the compiler chooses kernels by timing them and some reductions use atomics. A reproducibility floor is the standard deviation of a metric over identical reruns, with the same seed, configuration and code. Each rule below came from a defect in the project's record; Appendix C lists every floor.

Figure 10. Reproducibility floors per GPU, per artifact and per run length, and the confirmatory grid against its floor, one tab each, with its confound printed. On the figure, "card" is the GPU, "box" the machine, and M5.1e, M5.5 and M6.0 are studies keyed in Appendix C. "Traced" and "baseline" are the refactored and reference versions of the code, compared by per-field trajectory hashes ("field digests"); "floor-basis sd(Γ)" is the floor-based standard error of Γ (§8.4). The figure cannot show the contrasts that have no floor (§8.6). Status: point estimates over identical reruns, about ±40 % uncertain at four reruns · Interact: click a tab or press ← / →; hover or tab to a mark.



Static companion of Figure 10 for the printed PDF.

8.1 A bar without a floor is void

No acceptance threshold enters an analysis script without a measured floor for the quantity it grades, or an explicit UNDERPOWERED flag. A threshold below its instrument's noise

cannot pass under the null, so its output is void, and a void test voids a PASS as it voids a FAIL. The rule followed four bars set without floors (Appendix C). In one, the communication-denial test graded a condition with a 20 % threshold, chosen at implementation, against a quantity whose rerun nondeterminism is 27.2 %. Since the threshold sits below the noise, the condition is VOID; the observed 23.2 % exceeded the one and fell short of the other.

Floors are facts per metric and per GPU. Re-measuring the Medium floor moved completion's 2.75-fold and left survival's unchanged, 0.0129 against 0.0130. The two floor studies differ in GPU, code version and configuration, so the exhibit shows that one metric's floor can move while another's stays put; it cannot say which factor moved it. The floor is measured on the GPU that runs the grid.

Floors are facts per artifact. In a refactor of the environment code, the rerun floor measured on the reference version read zero, so two diagnostic channels that differed between the versions looked like a semantic change. The refactored version then differed from itself 1 time in 4 on the same channels. An equivalence claim between A and B needs A against A and B against B as well as A against B.

8.2 Contrasts and power

A contrast is graded on its own standard error. For Γ, the difference between the JOINT and ISO arm means over k_{JOINT} and k_{ISO} seeds with seed standard deviations s_{JOINT} and s_{ISO},

$$\mathrm{sd}(\Gamma) = \sqrt{\frac{s_{\mathrm{ISO}}^2}{k_{\mathrm{ISO}}} + \frac{s_{\mathrm{JOINT}}^2}{k_{\mathrm{JOINT}}}},$$

which for equal k is $\sqrt{(s_{\mathrm{ISO}}^2 + s_{\mathrm{JOINT}}^2)/k}$. The per-arm forms, each arm's own $s\sqrt{2/k}$, bracket that value without equalling it: at $k = 34$, ISO's gave 92.1 % power and JOINT's 62.8 %, where the contrast's power was 76.7 %.

A design-stage power statement is the minimum detectable effect at 80 % power and the family-corrected level,

$$\mathrm{MDE}_{80} = \left(z_{\alpha} + z_{0.8}\right)\ \mathrm{sd}(\Gamma),$$

with $z_{\alpha} = 2.2365$ the Šidák-corrected critical value for the two co-primaries. The exclusion bound X that §9 attaches to a null is this quantity on the grid's realized seed dispersion. A bare $2\sigma\sqrt{2/k}$ is the effect at which a test would just reject, which it detects with 50 % power; it survives only for grading an observed effect against a floor after the fact, labelled as such. The confirmatory seed count was first ruled at $k = 20$ on such a figure, without the Šidák correction; corrected, completion had 55.6 % power against its motivating effect band, a slip caught before any run.

8.3 Floors move with the run, and gates must not divide by them

Floors grow with run length. From 500 to 1000 updates the completion floor grew 2.1-fold for ISO and 5.2-fold for JOINT, with length confounded with machine. Run length is part of the artifact a floor describes.

A gate that divides by a floor diverges on more reproducible hardware. A registered convergence check, retired before the grid, divided each arm's late-training drift by its floor, so any nonzero drift fails as the floor goes to zero; it fired when JOINT's floor fell faster than its drift. The T^* ruling of 2026-08-11 retired it as a gate, fixed the estimand as Γ at matched compute, and made Γ(t) required robustness evidence (§9.2).

8.4 The floor is a check, not the test variance

The floor and the test statistic are different variances. A floor comes from identical reruns at one seed, while a k-seed arm mean varies by seed as well. The confirmatory contrast is graded on the grid's own seed dispersion. The floor serves the check that an effect exceeds rerun noise, which uses the floor-based standard error of Γ: the display in §8.2 with floors in place of seed standard deviations (§9.2). For design-stage power, the floor gives an upper bound, labelled as one.

8.5 Instruments state what they are blind to

A differenced test sees asymmetry and nothing else. A condition of the form "no cross-arm difference in X" cancels any X present in both arms. In 24 of 24 rendered episodes of the Coupling-B calibration grid, the agents drifted towards a wall (rendered-pixel centroids, a diagnostic). That grid's inertness falsifier, logged before its data, included the condition "no cross-arm exposure/positioning difference", which could not see a drift present in both arms; the verdict it fed rests on its other conditions.

Measuring is not grading. A recorded channel becomes evidence only inside a registered analysis family, with a floor measured on its artifact. Position-drift diagnostics added before the grid are recorded for every grid run. They are recorded, not graded: adding them to the registered family after registration would have enlarged it and inflated its correction. Bias (b) of §9.1 changed sign between the three floor studies that read it before unblinding; §9.1 reports every reading.

8.6 Where this paper falls short of its own rules

Coupling B's Medium survival effect was judged only against a three-seed spread (§6.4). The trained-policy contrasts of §5 and §6 come from two calibration grids without identical reruns: their spreads are seed ranges, and any floor shown beside them is a reference scale (cross-artifact). In the secondary design, floors exist only for the sweep arm at $p = 0.5$; the intermediate sweep arms assume that floor, and the identification arm is ungraded. ISO-4 has no identical reruns, so the contrasts that involve it are graded on seed dispersion alone, and the High-severity readout flags ISO-4 UNDERPOWERED for the check against rerun noise. The position-drift diagnostics have no floor. Every floor is itself a point estimate: measured on four identical runs, a standard deviation is uncertain by roughly ±40 %.

9 The pre-registered compositional test

9.1 Design

Definition 8 (Training protocols and the compositional gap). Let

$J_\theta(\pi)$ be the expected episodic return of joint policy π in $\mathcal{M}_\theta$, and let θ^* be the test configuration, every stressor on at a severity held out of training. Let Θ_{iso} be the configurations in which at most one of κ_A, κ_B, δ is nonzero; the fire ($\beta > 0$) burns in all of them. The ISO protocol trains one policy π_{iso} on a uniform mixture of configurations in Θ_{iso}. The JOINT protocol trains one policy π_{joint} on a mixture whose support includes configurations with every stressor on, at training severities only. The compositional gap is

$$\Gamma(\theta^*) = J_{\theta^*}(\pi_{\text{joint}}) - J_{\theta^*}(\pi_{\text{iso}}).$$

The registered test replaces J by an outcome metric's mean over evaluation episodes, on two co-primaries: task completion, the founding primary, on which the founding hypothesis was $\Gamma(\theta^*) > 0$, and survival rate (§3.5). Survival became a co-primary on 2026-08-02, before the grid, because both couplings' calibration grids had moved survival while their completion effects sat at or below the reproducibility floor (§5–§6).

Proposition 4 (A transfer bound for a fixed policy). For a fixed policy and two configurations that share the reward, the difference between its discounted returns is bounded above by a term that grows with how far their transition and observation kernels differ. Every configuration in Θ_{iso} differs from θ^* on the states where both couplings act at once, a collapse-seeded fire within perception-attenuation range of an agent (proof idea in Appendix A).

Proposition 4 bounds what ISO can be *guaranteed* to transfer, not what happens. It does not lower-bound Γ, and compositional generalization can emerge from isolated training; it motivates the experiment, and the experiment decides. Its prediction about ISO's exposure to that region was not tested, because the counter built to measure visits there (§6.5) is co-active only in all-on configurations (§9.2).

The test configuration θ* has every stressor on at the held-out severity, β 0.49, κ_A 0.06, κ_B 1.0, δ 1.0, and no arm trains at β 0.49 (Figure 2 in §1). In the table, μ is the share of training episodes carrying a given coupling and p the share carrying both. The active share is the share with at least one behaviourally active stressor, and the fire-only share v is the rest.

arm	training components	weight each	per-coupling frequency μ	co-occurrence p	active share	fire sh (be
ISO	{A-only, B-only, δ-only} × {0.43, 0.70}	1/6	1/3	0	2/3	1/3
ISO-4	{A-only, B-only} × {0.43, 0.70}	1/4	1/2	0	1	0
JOINT	{all-on} × {0.43, 0.70}	1/2	1	1	1	0

ISO-4, a sensitivity arm outside the confirmatory family, is ISO without its two δ-only components, and the corrected gap Γ_4 is JOINT − ISO-4. All arms share the network and the IPPO hyperparameters (Appendix B), one shared-parameter policy per training seed [11], a budget of T* = 1000 updates of 32,768 environment steps each, and one evaluation draw, seed 0 × 512 episodes at θ*.

Γ is read at matched compute, at the fixed budget T*, and convergence is claimed nowhere. The confirmatory family is the two co-primary tests, Šidák-corrected (m = 2): unpaired normal z tests against $z_\alpha = 2.2365$ on the contrast's standard error from each arm's seed dispersion (§8.2). A null carries its exclusion bound X, the 80 %-power MDE at that α. Positive cross-arm covariance, the expected sign given the shared seed integers, makes the unpaired test conservative. Floors serve only the check that an effect exceeds rerun noise (§8.4) and, as an upper bound, design-stage power (Appendix D); every interval is conditional on the one evaluation draw.

The design registered k = 40 seeds per arm and a seed rule, capped at 60, that recomputes k from floors re-measured on the grid's GPU. It gave 46 on an RTX PRO 6000 and 72 on GPU 1, the RTX 5090 that ran the grid (GPU 2 and GPU 3 are two more of that model; Appendix F). Raising the cap to 72 on 2026-09-22 was a deviation from a registered rule, recorded as one, made before unblinding, and conservative, since added seeds favour neither sign of Γ. The registered k = 60 analysis, on seeds 1–60, is reported beside it; realized completion power at a 0.03 effect is 60.2 % and 71.6 % at k = 60 and 72.

JOINT sees each coupling three times as often as ISO, so Γ mixes per-coupling frequency with composition. A secondary sweep holds μ = 0.5 while p moves from 0 to 0.5, in the spirit of DBCA [12], which the ISO and JOINT endpoints do not follow. On the triangle of training mixtures $\mu = (1 + p - \nu)/2$, so a fixed μ forces ν = p. Identification arms on the parallel line μ = 0.4, ν = 0.2 + p identify a plane in p and ν (Appendix H). Secondary arms run k = 20 seeds with δ off, so the p = 0 sweep arm trains on ISO-4's mixture on JOINT's GPU, and JOINT, unlike every secondary arm, carries δ = 1.0, inert (§7). The secondary design is **non-verdict-bearing**.

The registered design contains an error, and the bias it creates inflates Γ. Communication denial had been found inert on 2026-07-30 (§7). The design registered on 2026-08-02 kept δ in θ* for fidelity to the registration and excluded it from the secondary sweeps because it was inert. It still gave ISO two δ-only components of six. Its review question on δ judged the arms symmetric, on the premise that neither trains on δ = 1.0, which is false for both, and asked for a second opinion. So a third of ISO's training episodes are fire-only, which the design never counted: ISO's configured no-stressor share of 0 is true nominally and false behaviourally, and JOINT's active share is 1.5 times ISO's. A review of the registered design identified this on 2026-08-16, before the grid, noting that the confirmatory arm applied a different rule to the same fact as the sweeps. The grid began on 2026-09-09 without the correction. The bias, its direction and ISO-4 with its reading rule were registered on 2026-09-18, mid-grid (120 of 252 runs) and before unblinding, on the owner's attestation that no outcome mean had been computed or viewed. The composition actually tested is A × B.

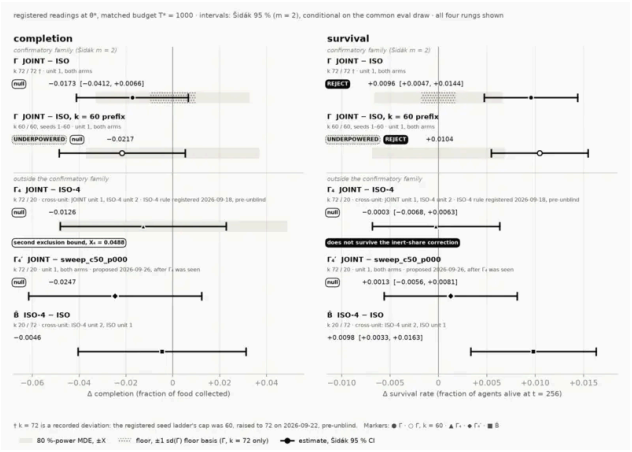
Bias (b), the second registered caveat, is differential drift: JOINT − ISO completion drift over the last 100 updates, a local sensitivity estimate and never a bound. It read +0.01209 per 100 updates on the first floor study (2026-08-10) and +0.0179 on the 2026-09-08 re-measured floor, both on an RTX PRO 6000, but −0.0083 on GPU 1. The T = 2000 subsample's differential training-log slopes over updates 500–1000 are +0.00247 (completion) and +0.00718 (survival) per 100 updates. The sign reverses on the grid GPU, and no direction is claimed.

The registered outcomes, branches A–D of 2026-08-13 and a fifth cell, A_c, added before unblinding, are in Appendix E; rejection at z_α on each co-primary fixes which outcome holds, and branch B (survival only) also needs a registered falsifier to pass. The ISO-4 rule states that if Γ rejects on a co-primary and "**Γ_4's CI on that co-primary includes 0 or its sign is opposite**, the paper states that the effect **does not survive the inert-share correction**". If Γ and Γ_4 are both null, "Γ_4 is quoted as a second exclusion bound". Appendix D quotes every rule with its date.

The grid, which ended on 2026-09-22, ran under a runner that computes no cross-arm quantity. The pre-unblind rulings of 2026-09-26 fixed the statistic, the sign rule, k = 72 as primary and A_c, and unblinding, the first computation of any outcome contrast, followed that day (Appendix F).

9.2 Results

Figure 11. The registered readings in order, one row (a "rung") per contrast: Γ, with the registered k = 60 analysis (labelled "prefix"); Γ_4; Γ_4', JOINT minus the p = 0 sweep arm (labelled sweep_c50_p000); and $\hat{B}$, which carries no verdict. "Unit n" is GPU n, "cross-unit" means across GPUs, and the "seed ladder" is the registered seed rule (§9.1). The grey band is the 80 %-power MDE; the stipple, on Γ only, is ±1 floor-based standard error of Γ (labelled "floor basis"). The figure cannot show effects present in both arms, evaluation-set sampling, other budgets, or which of active share and per-coupling frequency carries ISO-4's rise. Status: registered (Γ); sensitivity arm (Γ_4); secondary (Γ_4') · Interact: step rows with ◀ ▶ or the arrow keys; toggle the bands; "table view".



Static companion of Figure 11 for the printed PDF.

At k = 72, Γ rejects on survival and is null on completion (Figure 11 and the contrast table). The completion null excludes a joint-training gain larger than 0.66 points. Its bound X, the gap of either sign the design would detect with 80 % power, is 3.29 points, and the negative point estimate does not reject. The registered k = 60 analysis reads the same and is flagged UNDERPOWERED.

All contrasts at θ*, T* = 1000, in points (rate × 100), with Šidák-corrected 95 % intervals unless marked; k is given in the order of the contrast's two arms.

contrast	k	completion	survival	status
Γ = JOINT − ISO	72 / 72	null: −1.73 [−4.12, +0.66], z −1.62	REJECT: +0.96 [+0.47, +1.44], z +4.43	confirmatory
Γ, registered k = 60 analysis	60 / 60	null: −2.17 [−4.86, +0.52], z −1.80	REJECT: +1.04 [+0.54, +1.55], z +4.67	seeds 1–60, UNDERPOWERED

contrast	k	completion	survival	status
Γ_4 = JOINT − ISO-4	72 / 20	null: −1.26 [−4.81, +2.28], z −0.80; second exclusion bound	null: −0.03 [−0.68, +0.63], z −0.09	sensitivity arm, across GPUs
Γ_4' = JOINT − the p = 0 sweep arm	72 / 20	null: −2.47 [−6.16, +1.22], z −1.50	null: +0.13 [−0.56, +0.81], z +0.41	secondary, same GPU, read at z_α
$\hat{\mathrm{B}}$ = ISO-4 − ISO	20 / 72	−0.46 [−4.05, +3.12]	+0.98 [+0.33, +1.63]	no registered test, no verdict; across GPUs
replicate difference: p = 0 sweep arm − ISO-4	20 / 20	+1.21 [−2.79, +5.20], 95 %	−0.15 [−0.86, +0.56], 95 %	secondary, across GPUs
$\mathbf{\Gamma}_H$, all-on at β = 0.70	72 / 72	null: −0.79 [−3.26, +1.68]	null: +0.38 [−1.46, +2.23]	evaluation only, out of family, not held out
$\mathbf{\Gamma}_{H,4}$ = JOINT − ISO-4 at β = 0.70	72 / 20	null: −0.75 [−3.87, +2.36]	null: −1.28 [−3.50, +0.95]	no reading rule of its own
Γ(2000)	4 / 4	+0.03 [−3.63, +3.69], 95 %	+1.59 [−0.02, +3.19], 95 %	descriptive, UNDERPOWERED

The registered falsifier passes: the completion contrast excludes, in standardized units, an effect as large as the observed survival effect, $|z_c| + z_\alpha = 3.855 < 4.433 = |z_s|$ (4.040 < 4.673 at k = 60). Branch B stands, completion null and survival positive, and branch C does not apply. Under Definition 2 the asymmetry is between a penalized channel and a rewarded one: the reward prices every death at $d_p = 0.5$ whatever its cause and never reads the hazard. Branch A, the founding hypothesis, did not occur, and neither did A_c.

Γ_4, the corrected gap JOINT − ISO-4, is null on survival. By the registered rule, Γ rejects and Γ_4''s interval contains 0, so the survival effect **does not survive the inert-share correction**, and this paper does not call it a composition effect. The branch label stays B, because the rule fixes it by Γ. Branch B's registered headline, "composition is paid in agents, not task return", no longer holds as worded. On completion both contrasts are null, so Γ_4 is a **second exclusion bound**, X_4 = 0.0488 (4.88 points).

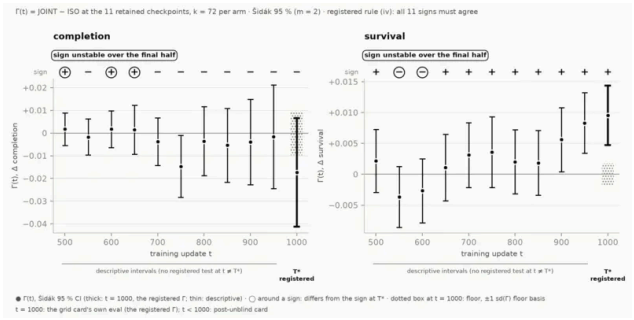
Γ_4 is a contrast across GPUs: ISO-4 trained on GPU 2, JOINT on GPU 1. Γ_4', the same-GPU replicate of the correction, is JOINT minus the p = 0 sweep arm, which trains on ISO-4's mixture on JOINT's GPU. Proposed on 2026-09-26 after Γ_4 was seen, before that arm was read, and unable by rule to change Γ_4''s qualifier, it agrees: null on both co-primaries.

$\hat{\mathrm{B}}$, ISO-4 − ISO, has no registered test and no verdict, and no magnitude threshold on it was registered. Because Γ = Γ_4 + $\hat{\mathrm{B}}$ holds identically for arm means, a $\hat{\mathrm{B}}$ close to Γ restates a Γ_4 close to 0 and is not a second finding.

Γ_4's upper bound, +0.63 points, lies above Γ's lower bound, +0.47, so the correction excludes a large composition effect but leaves a smaller one possible. Removing the δ-only episodes also raises both the active share, from two thirds to one, and each coupling's frequency, from a third to a half, and $\hat{B}$ alone cannot say which matters. What Γ_4 does say is that, once the active share is matched, JOINT's remaining differences add nothing detectable to survival, although JOINT still sees each coupling twice as often as ISO-4 and always in combination.

$\Gamma(t)$ is **sign unstable over the final half** on both co-primaries (Figure 12): the registered rule calls the result budget-robust only if all 11 point estimates over updates 500–1000 share one sign. As registered, the instability is the finding, and every reading here holds at matched compute only. Survival $\Gamma(t)$ stays between −0.37 and +0.36 points through t = 850, every interval containing zero. It reads +0.56, +0.83 and +0.96 points at t = 900, 950 and 1000, the only checkpoints whose Šidák interval excludes zero: the survival gap opens only in the last fifth of the budget. Completion is negative at every checkpoint from t = 700, 70 % of the budget, and its interval excludes zero only at t = 750, −1.47 points [−2.83, −0.10], descriptively. Completion is unresolved and leans negative.

Figure 12. Sign unstable over the final half, on both co-primaries. $\Gamma(t)$ is JOINT − ISO over the final half of training, with Šidák whiskers. A ringed sign differs from the sign at T*; rule (iv) is the pre-unblind sign ruling quoted in Appendix D. Only the checkpoint at T* is the registered Γ; earlier intervals are descriptive. "Grid card" is GPU 1 and "post-unblind card" GPU 3; the "single-card curve" re-evaluates the T* checkpoints on GPU 3. The dotted box is ±1 floor-based standard error of Γ (labelled "floor basis"). The figure cannot show where either difference goes after T*. Status: registered at T*, descriptive before · Interact: hover a checkpoint; toggle the single-card curve and the box.

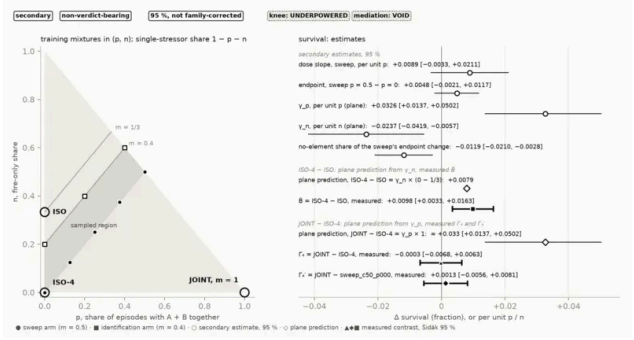


Static companion of Figure 12 for the printed PDF.

The T = 2000 subsample (4 seeds per arm, UNDERPOWERED) has positive differential training-log slopes over updates 1000–2000, +0.00204 (survival) and +0.00384 (completion) per 100 updates. The drift has not visibly reversed, which is all it may be cited for.

The secondary design (Figure 13; Appendix H) reports 95 % intervals that are not family-corrected. At ISO's fire-only share of 1/3, the survival plane's fire-only-share slope γ_n predicts ISO-4 − ISO = +0.79 points, inside $\hat{B}$'s interval. This does not attribute $\hat{B}$ to the fire-only share as such, because at fixed p a move in v also moves μ. The co-occurrence slope γ_p is positive across the sampled region, but moving single-stressor into both-stressor episodes also raises each coupling's frequency, and the triangle does not separate the two. Fitted on p up to 0.5, the plane predicts JOINT − ISO-4 = +3.26 points at p = 1, far above the measured Γ_4 and Γ_4': γ_p does not extrapolate to JOINT.

Figure 13. The secondary design; the change point is labelled "knee". Left: the triangle of training mixtures in co-occurrence p and fire-only share v (labelled n), lines of fixed per-coupling frequency μ (labelled m), and the arms. Right: the secondary estimates, where γ_p and γ_n are the co-occurrence and fire-only-share slopes, the "dose slope" is the sweep's slope in p, and the "no-element share" is the fire-only share's part of the sweep's endpoint change. Below, the plane's γ_n prediction sits beside $\hat{B}$, which it fits, and its γ_p prediction beside Γ_4 and Γ_4', which it misses; the p = 0 sweep arm is labelled sweep_c50_p000. The figure cannot separate per-coupling frequency, co-occurrence and active share. Status: secondary · Interact: switch the metric; toggle the predictions; hover.



Static companion of Figure 13 for the printed PDF.

On a secondary metric outside the family, descriptive and without a floor, JOINT has 0.1177 fewer fire deaths per episode than ISO (z = −4.59); expressed in agents, the survival gap is about 0.11 per episode of 12.

The High-severity readout re-evaluated every T* checkpoint of ISO, JOINT and ISO-4 with every stressor on at β = 0.70, out of family. Both co-primaries are null, so the reading is an **a-fortiori exclusion**: no gap larger than 3.40 completion points or 2.54 survival points at High even where JOINT trained on the evaluated configuration and ISO never saw a composed one. Γ_H lies within rerun noise, at 0.73 and 0.75 times its floor-based standard error; ISO-4 has no identical reruns, so that check is UNDERPOWERED for it. The readout is not a held-out test: JOINT trained on this configuration at weight 1/2. At High, Coupling B's survival effect replicated in direction in 3 of 3 contrasts (§6.4) while Coupling A is marginal (§5.3), the mirror image of θ*'s Medium siting (§10).

Mediation is **VOID**. The registered mediator, the counter's realized count during training, had a first-stage slope on p of -1.03×10^{-5}, 95 % $[-2.91 \times 10^{-5}, +0.93 \times 10^{-5}]$. A rule fixed before the sweep was read voids it when that interval contains zero. Diagnosed after the run, with the rule unchanged: the counter counts collapse-seeded ignitions inside an alive agent's crop, which only Coupling A produces. In mixtures it measures the Coupling-A frequency that the sweep holds fixed and is blind to co-occurrence.

Seed dispersion exceeds the rerun floor in both arms, 1.12 and 1.09 times on completion and 1.25 and 1.19 times on survival (ISO, JOINT). Survival Γ, at 5.40 times its floor-based standard error (completion 1.78 times), clears the check that an effect exceeds rerun noise. Γ and Γ_4' ran on GPU 1, ISO-4 and the T = 2000 subsample on GPU 2, and the post-unblind evaluations on GPU 3 (Appendix F); §11 reports how closely evaluation reproduces on one GPU and across GPUs.

9.3 A recommendation and its scope

A comparison of joint and isolated training that leaves the active share free can report an active-share effect as a composition effect. This holds by construction, as JOINT's threefold per-coupling frequency over ISO does. We make one recommendation, motivated by this result alone: that studies of compositional generalization match the share of training that contains an active stressor before attributing a gain to composition. It was motivated in one environment at one scale, and whether it holds elsewhere is untested. Matching the active share here left JOINT's other differences in place, and the corrected contrast excludes only a large composition effect (§9.2).

10 Limitations

Items 1 to 10 are properties of the design, stated whatever the outcome. Each was listed, in this order, before unblinding, and item 1 was registered as applying to every outcome.

1. θ* sits at Medium, where Coupling B's own measured survival effect was −0.03 points, graded only against a three-seed spread. Its survival effect is at High, where the Coupling-B contrast replicated in direction three times out of three, between −4.7 and −10.7 points; the registered text quotes the first two-seed estimate, 8.8 points. Medium was chosen because both couplings clear their lock criteria there and its floors are the smallest, which trades the existence of a behavioural effect for measurability. By the project's own locks, no severity has both couplings strongly live: at Medium B is quiet and A live, and at High A is marginal by construction (§5.3) and B live. The High readout adds breadth in severity and does not test composition where both couplings bite. The pre-unblind line that read a Medium survival effect as one "the single coupling did not show" is struck: Γ_4 and Γ_4' contradict it.

2. θ* is one held-out point, interpolated in one scalar between the training severities β = 0.43 and 0.70, and no training data lies near criticality.

3. The task requires no coordination: twelve agents share one policy's parameters and a team reward of +1 per collected item; "swarm" is not a claim.

4. The reward sees deaths, at $d_p = 0.5$ each, and not the hazard, so the asymmetry between survival and completion is between a penalized channel and a rewarded one. The only d_p ablation is narrow and was not re-run under the protocol of §9 (§3.3).

5. Communication denial is inert under a frozen random-projection encoder at this scale: the message head receives no gradient, and gradient-shaped messaging remains untested. The composition the test probes is A × B; δ is carried in θ* for registration fidelity.

6. The observation includes the agent's own position and the time step, so a policy can condition on where and when it is, independently of the hazard. The wall-drift artifact (§8.5) is the visible form, and no differenced test can see it; the position-drift diagnostics are recorded, not graded.

7. Every interval in §9 is conditional on one common evaluation draw, seed 0 × 512 episodes, which cancels in Γ; the intervals carry training-seed variance only.

8. The confirmatory runs and every analysed secondary run, the same-GPU replicate's included, trained on GPU 1; ISO-4 trained on GPU 2, so Γ_4 and $\hat{\mathrm{B}}$ are across GPUs; and the post-unblind evaluations ran on GPU 3 (Appendix F).

9. No external baseline, domain-randomization schedule or adaptive curriculum is evaluated; every arm uses one learner, single-policy IPPO, with one hyperparameter set (Appendix B).

10. The hazard is simple: a one-step burn time, a single ignition at reset, no spontaneous ignition ($\iota = 0$), and no wind or fuel heterogeneity. Corollary 1's exact critical value is specific to this idealized kernel on the infinite lattice.

Registered or recorded before unblinding:

- The confirmatory comparison carries a design error. The registered design kept δ-only episodes in ISO's training mixture after the communication axis had been closed as inert, and left δ out of the secondary sweeps for that reason. It did not count the resulting fire-only share, a third of ISO's training episodes, and the bias inflates Γ. Its correction was registered before unblinding but rests on a 20-seed sensitivity arm outside the confirmatory family, trained on another GPU, and the branch label stays B (§9.1).
- Several bars have no floor on their own artifact, because the design ran no identical reruns for them (§8.6).
- The confirmatory seed count exceeds the registered seed rule's cap, a deviation recorded before unblinding (§9.1).
- Proposition 4 bounds what isolated training can be guaranteed to transfer and does not lower-bound the realized gap; the experiment decides (§9.1).

Found after unblinding:

- The survival effect does not survive the inert-share correction, which excludes only a large composition effect and leaves open whether the active share or the per-coupling frequency carries ISO-4's rise over ISO. The same-GPU replicate, proposed after Γ_4 was seen, agrees (§9.2).
- On both co-primaries Γ(t) is sign unstable over the final half, so Γ is a claim at matched compute only; the T = 2000 subsample, UNDERPOWERED by registration, cannot settle another budget (§9.2).
- Completion is unresolved: its null is an exclusion with its bound, and it leans negative late in training (§9.2).
- The registered mediator was blind to co-occurrence in mixed training, so mediation is VOID (§9.2).
- The secondary analyses report many intervals, none family-corrected (Appendix H).

All of it was measured in one environment, at one scale, with one learner: the scope of the recommendation in §9.3.

11 Reproducibility

The code behind this paper is one repository. It holds the environment, training, evaluation and calibration code, and every configuration, θ* and the generated protocol configurations included. It also holds the analysis scripts that produced the results of §9, a registry of the locked constants, and a CPU test suite that runs the theory checks as unit tests. A test asserts that the configurations and code defaults agree with the registry and that each registered analysis constant equals the literal in the script that uses it. Each listing in this paper is a verbatim excerpt of the repository, labelled with its file and lines.

Ablations are exact by construction: every stochastic branch consumes its random numbers whether or not its stressor is on, so switching a stressor off reproduces the nested model's trajectories bit for bit under the same keys (§3.4, Listing 3). Four repeated evaluations of each of two checkpoints are bit-identical on one GPU and identical to the grid's own evaluations. Across GPUs, 280 of 288 paired evaluations are identical, and the largest absolute difference is 2.44×10^{-4}. GPU training is not run-to-run reproducible (§8), which is why floors are measured (Appendix C).

On GPU 1, an RTX 5090, a 1000-update training run took 518–521 s and a 512-episode evaluation 9–10 s (Python 3.12.3, jax 0.11.0). A run is 32,768,000 environment steps, each advancing all 12 agents; Appendix B gives the hyperparameters. These are training measurements. The paper states no environment-only throughput figure, because under the compiler's dead-code elimination such a figure depends on which outputs its consumer reads.

The checkpoints of the experiment in §9 were archived off the rented machines with recorded sha256 hashes, except a partial floor set that grades nothing, in which one of 12 archives has a hash (Appendix F). A re-verification matched every recorded hash. The analysis scripts refuse to run out of order, and each stage's log records a single run (§9.1).

Appendices

- Appendix A: proofs.
- Appendix B: locks and hyperparameters.
- Appendix C: reproducibility floors.
- Appendix D: the frozen analysis plan, quoted.
- Appendix E: the registered outcome branches.
- Appendix F: grid provenance and blinding.
- Appendix G: Γ over the final half of training.
- Appendix H: the secondary design.
- Appendix I: trained-policy episodes.

A Proofs

Each statement is in the body, in the section named. The proofs follow the formal specification in the notation of Definition 1, and departures from it are noted.

Proposition 1. Statement in §3.1.

Proof. Each factor of T_θ is a conditional kernel, measurable in its conditioning variables, all of which are components of (s, a) or of already-sampled components of s'. The dependence of T_H on c' and of T_K on x' is a sequential composition within one time step. Sample $c' \sim T_C(\cdot \mid c, x)$, then $h' \sim T_H(\cdot \mid h, c, c')$, then $\rho' = T_\rho(\rho, h')$ (deterministic), then $x' \sim T_X(\cdot \mid x, a, h', c')$, then $k' \sim T_K(\cdot \mid x')$. A composition of Markov kernels is a Markov kernel on the product space, so T_θ is a valid transition kernel on $\mathcal{S}$ and (s_t) is Markov. The action sets are finite, every stochastic factor and the reset distribution have finite support, and ρ evolves deterministically from $\rho_0 = 0$. The states and observations reachable within the horizon therefore form finite sets; restricted to them, $\mathcal{M}_\theta$ is a finite Dec-POMDP, and optimal finite-horizon joint policies exist by the standard argument. ■

The formal specification states that $\mathcal{S}$ and Ω_i are finite. With the real-valued smoke field $\rho \in \mathbb{R}^G_{\geq 0}$ in the state, that holds only for the states and observations reachable within the horizon, which is all the argument uses.

Proposition 2. Statement in §3.2.

Proof (coupling). Attach to each unordered edge $e = \{g, g'\}$ an independent $B_e \sim \mathrm{Bernoulli}(\beta)$, and declare that a transmission attempt across e succeeds if and only if $B_e = 1$. Each edge is attempted at most once: when one cell of e first Burns while the other is Fuel, one attempt occurs; afterwards the burning endpoint is Burnt, which is absorbing. If the other endpoint later Burns via a different path, its neighbour across e is already Burnt, so no second attempt occurs. Run with the variables $\{B_e\}$, the fire therefore burns exactly the connected component of the origin in the subgraph of open edges $\{e : B_e = 1\}$, which is the bond-percolation cluster. ■

This is the coupling behind the correspondence that §3.2 cites [9], and the implementation realizes it exactly (`che/env/hazard.py:8-16`). It draws one uniform per cell and incoming direction, which maps one-to-one to the ordered pairs of Definition 3. With a burn time of one step, each unordered edge sees at most one attempt, so these uniforms are the coupling's per-edge variables.

Proposition 3. Statement in §5.2.

Proof. Each collapse seed grows, by Proposition 2, a burnt cluster equal in distribution to a bond-percolation cluster, of expected size $\chi(\beta)$. By linearity of expectation over the Binomial or Poisson number of seeds, the total expected burnt area is $\lambda_A T \chi(\beta)$ when clusters are disjoint; overlaps only remove double counting, which gives the upper bound in general. ■

The formal specification adds a divergence rate taken from Corollary 1(3), which the paper does not state (§4.1). The equality needs four conditions matched (cluster-mass reference, seed location, seed age and sparsity), and cluster mass at β = 0.43 saturates by an age of about 64 steps. In the Proposition 3 measurements (§5.2) the four factors compose to the observed slope within Monte Carlo error.

Theorem 1. Statement in §6.2; items (1)–(3) below refer to it.

Proof. *(1)* With zero slack, the only decision is the corridor choice at b; waiting or probing is impossible within T. If informed (probability q), taking the corridor other than Z succeeds with probability 1; if not, both corridors are exchangeable given the history, and any choice succeeds with probability $\frac{1}{2}$. Hence $J^* = q + (1-q)\frac{1}{2} = \frac{1}{2} + \frac{q}{2}$, and no policy exceeds this, since success requires information or a lucky guess. *(2)* In the fixed-map environment the signal is uninformative, so the signal-blind policy is among the optimal ones there; on E_{2C} it succeeds if and only if $Z = z_0$, with probability $\frac{1}{2}$. *(3)* Each factor $\left(1 - e^{-\kappa_B j}\right)$ is continuous and strictly increasing in κ_B on $(0, \infty)$, so the finite product is too, and q is continuous and strictly decreasing. The product tends to 0 as $\kappa_B \to 0$ and to 1 as $\kappa_B \to \infty$. ■

Proposition 4. Statement in §9.1.

Proof idea. The difference between the two returns is bounded by telescoping the evaluation operators of the two models along the trajectory distribution; the observation term enters because the policy acts through the observation. Every configuration in Θ_{iso} differs from θ^* where both couplings act at once. Take a state with a collapse increment next to Fuel and an agent within attenuation range. There T_{θ^*} places mass of order κ_A on ignition transitions that a configuration with $\kappa_A = 0$ places elsewhere, and the observation kernels differ when κ_B differs.

The formal specification displays the bound with an explicit constant and concludes more about ISO's slack than an upper bound can show; the paper states the bound in words and draws no such conclusion (§9.1).

Remark 2'. Stated in §7.2, after Pynadath and Tambe[35]. One step is not spelled out in §7. Let $\tilde{q}$ be the probability of being informed under the best idling schedule. The commitment draws have different optical depths, so as $\kappa_B \to \infty$ the lowest-depth draw dominates both products and $\tilde{q}/q \to 1$. The correction peaks at moderate κ_B, and its relative bite on the value of communication is largest at low κ_B, where q is near 1. In the two-agent toy, which is not CHE (4096 episodes and 8192 Monte Carlo draws per point), $\tilde{q}/q$ is 1.126 at $\kappa_B = 1.0$ and peaks at 1.235 near $\kappa_B = 2$. The correction removes 79 %, 54 % and 7 % of $\mathrm{VoC}_{\mathrm{gated}}$ at κ_B = 0.5, 1.0 and 3.0. The handshake gate on the pinned arm passes (largest absolute z 0.44; χ^2 0.51 on 8 degrees of freedom), and the free arm scores 1.0000 at every κ_B.

B Locks and hyperparameters

Each locked constant below was fixed by a ruling on the date given, and the test described in §11 checks each against the configurations, the code defaults or the analysis scripts.

constant	value	ruled	record
critical point $\hat{\beta}_c$	0.500 ± 0.005	2026-07-19	severity lock record; no committed script reproduces its estimator, and the committed estimators give a crossing mean of 0.504 and a consensus of 0.499
β Low / Medium / High	0.43 / 0.49 / 0.70	2026-07-19	severity lock record
β held out (θ*)	0.49	2026-08-02	the ruling that sited θ*: train at the two extremes and hold out the middle
f_{weak}, λ_0, λ_{load}, κ_A, r_A	0.15, 5e-5, 4e-4, 0.06, 1	2026-07-21	Coupling-A lock record
weak-floor smoothing passes	2	2026-07-21	Coupling-A lock record, locked as the code default
κ_B	1.0	2026-07-27	Coupling-B lock record, on the detection band alone
δ off / on; R_{comm}	0.0 / 1.0; 16	2026-07-30	communication lock record
death penalty d_p	0.5	2026-07-19	the death-penalty design decision
observation encoding	3, the masking observation (§3)	2026-07-24	the observation design decision
σ_s, η, ι	1.0, 0.5, 0.0	not locked	held fixed by convention, never calibrated

analysis constant	value	ruled	role
`K_CONFIRMATORY`	40	2026-08-03	the design seed count per confirmatory arm, against which the registered seed rule computes

analysis constant	value	ruled	role
`K_LADDER_CAP`	60	2026-08-03	the seed rule's cap
`K_CONFIRMATORY_REALIZED`	72	2026-09-22	the seed count at which Γ is read, a recorded deviation (§9.1)
`K_SECONDARY`	20	2026-08-02	seeds per secondary arm; ISO-4 uses the same count
`SIDAK_M`; z_α	2; 2.2365	2026-08-02	the confirmatory family: the two co-primary tests, Šidák-corrected
`POWER_STOP`	0.80	2026-08-02	the minimum design-stage power at the 0.03 band
`T_STAR`	1000	2026-08-11	the fixed training budget in updates, a value unchanged since 2026-08-03
`GAMMA_T_RETENTION`	11	2026-08-11	checkpoints kept for Γ(t): updates 500–1000 every 50
evaluation draw	seed 0 × 512 episodes	registered with the grid layout	common to every run; not in the lock registry
`PLATEAU_PASS`; `PLATEAU_REVIEW`	1.0; 1.5	2026-08-02	thresholds of the registered convergence check, retired as a gate on 2026-08-11; they grade nothing

The training hyperparameters are identical for every run of §9; the grid passes none on the command line, so the configured defaults apply.

hyperparameter	value
learning rate	3e-4, constant, no schedule
discount γ; GAE λ	0.99; 0.95
PPO clip ε	0.2
entropy coefficient; value coefficient	0.01; 0.5
global gradient-norm clip; Adam ε	0.5; 1e-5

hyperparameter	value
parallel environments; steps per update	256; 128
minibatches; epochs	4; 4
checkpoint interval	50 updates
budget	32,768 environment steps per update, 32,768,000 per run

C Floors

A floor is the standard deviation of a metric over identical reruns at one seed, so it measures nondeterminism alone. Floors are per metric, per GPU, per artifact and per run length; the table lists each one this paper uses, under its study code (key below). With four reruns a standard deviation has three degrees of freedom and is uncertain by about ±40 %, and the confirmatory training floors use eight.

study	GPU	updates	artifact	reruns
M5.1e	RTX 5090	500	Medium, couplings locked, communication live at δ = 0	4
M5.5	RTX PRO 6000	500	Medium, R_{comm} 16	4
M5.5, danger moments	RTX PRO 6000	500	Medium	4
M5.3b	RTX PRO 6000	500	High	4
M6.0	RTX PRO 6000	n/a	per-field trajectory hashes, refactored against reference version of the code	4 comparison
M6.2	RTX PRO 6000	500	ISO, JOINT	8
M6.2b	RTX PRO 6000, another machine	1000	ISO, JOINT	8

study	GPU	updates	artifact	reruns
G1.2, 2026-08-10	RTX PRO 6000	1000	ISO, JOINT, sweep arm at p = 0.5	8
re-measured, 2026-09-08	RTX PRO 6000	1000	ISO, JOINT, sweep arm at p = 0.5	8
re-measured on GPU 1	RTX 5090	1000	ISO, JOINT, sweep arm at p = 0.5	8
evaluation floor	RTX 5090, GPU 3	evaluation only	two checkpoints	4
partial re-measurement, 2026-08-14	not recorded	n/a	ISO, JOINT	8 ISO + 4 JOINT evaluated

The floors of GPU 1 cannot be distinguished from the RTX PRO 6000's: the 95 % intervals for the standard deviation overlap (ISO completion [0.0186, 0.0571] on the RTX PRO 6000 against [0.0388, 0.1194] on the RTX 5090). No claim is made that either GPU is less reproducible.

The rule of §8.1 followed four bars set without floors. The Coupling-B calibration grid graded its High survival effect against a two-seed spread of 0.0295, below the High floor of row M5.3b, and its "strong" grade was retracted (§6.4). The message-use test at Medium applied a floor hard-coded from the study of row M5.1e. Re-sited at High, it measured its floor first and found that matching Medium's resolution would take roughly 46 seeds per arm. The communication-denial test set its third condition's threshold below the rerun nondeterminism of row M5.5, danger moments. Those floor data predate the comparison, and the ruling records that a difference above the floor would have left the verdict NOT INERT standing. In the per-artifact exhibit, both versions of the code ran on one GPU, and the two channels that differed were masking diagnostics at High severity.

code	plain name in the text
M3.5 (printed m35)	the Coupling-A calibration grid
M4.4 (printed m44)	the Coupling-B calibration grid
M5.1e	the four-run Medium floor study on an RTX 5090
M5.3, M5.3b	the message-use test at Medium, and the same test re-sited at High with its own floor study
M5.5	the communication-denial test, with its Medium floor study on an RTX PRO 6000
M6.0	the refactor that turned the environment configuration into data, and its rerun comparisons
M6.2, M6.2b	the confirmatory arms' floor studies at 500 and 1000 updates
G1.2	the grid's launch floor study

D The frozen analysis plan

The registered constants are in Appendix B. The reading rules follow, quoted from the registration record with their dates: emphasis is the record's own, one indented code line is set inline, and omissions are marked […]. A "§" inside a quotation refers to the record's own sections, and S1–S6 are its labels for the secondary analyses.

1. The estimand (T* ruling, item 2, 2026-08-11):

> ***Primary Γ = the confirmatory contrast at MATCHED BUDGET T = 1000, all arms — registered as a fixed-budget claim.***
>
> ***"At convergence" is explicitly NOT claimed anywhere.*** *The paper says* ***"at matched compute"*** *throughout, which is also the practitioner-relevant question.*

2. Budget robustness (item 3 of the same ruling; restated for every branch on 2026-08-13):

> *The fixed-budget conclusion is reported as* ***budget-robust iff the sign of Γ is stable over the final half of training****. If the sign is still moving at T,* ***that instability IS the finding*** *and is reported as such.*

> *Mechanism: 11 retained checkpoints, updates 500–1000 at interval 50, confirmatory arms only,* ***evaluated post-unblind****. Every branch above gains "— budget-robust" or "— sign unstable over the final half" as a suffix, and the suffix is not optional.*

3. The branch-B falsifier (2026-08-13):

> *Branch B is claimable* ***only if*** *the completion contrast excludes an effect as large, in standardized units, as the observed survival effect:*
>
> `|z_c| + z_α < |z_s|, z = Γ / sd(Γ) per metric`
>
> *with* `sd(Γ)` *the grid's own measured per-arm seed dispersion in the combined form and* `z_α = 2.2365` *the two-sided Šidák critical value at m = 2.* ***If it fails, the result is branch C with an asymmetry the design could not resolve — not branch B.***

4. Test variance and floor (the frozen analysis plan, amended 2026-08-13):

> ***The confirmatory tests and CIs are computed on the GRID'S OWN measured per-arm seed dispersion, not on the floors.*** *The floors are 8 identical reps at one seed; the grid averages over k distinct seeds, and those are different variances. The floors keep exactly two narrower roles: the* ***beat-reproducibility hurdle****, and a* ***design-stage power basis registered as an UPPER BOUND*** *(§5, superseded note, item 2).*

Design-stage power is therefore a ceiling. The first floor study (an RTX PRO 6000, 2026-08-10) gave, at k = 40, an MDE_{80} of 1.55 completion and 0.95 survival points; the grid GPU's floors gave completion power at a 0.03 effect of 72.2 % at k = 60 and 80.4 % at k = 72.

5. The inert-share rule (2026-09-18):

- *Γ and Γ_4 are reported **side by side on every branch**, with $\hat{B}$ and its CI.*
- *If Γ rejects on a co-primary and **Γ_4's CI on that co-primary includes 0 or its sign is opposite**, the paper states that the effect **does not survive the inert-share correction**, in the abstract-level summary and not only in an appendix. **The branch label does not change** — it is fixed by Γ — but branch A/B text may not describe the effect as a composition effect without that qualifier.*
- *If Γ is null and Γ_4 is null, Γ_4 is quoted as a second exclusion bound.*
- *If Γ is null and Γ_4 rejects **negative**, it is reported under branch D's framing (matched-budget asymmetry) as a secondary observation, never as a confirmatory finding.*
- ***No magnitude threshold on $\hat{B}$ is introduced**, now or later.*

6. The High readout, its reading for a null (same ruling):

- ***Γ_H null → an a fortiori exclusion**, and the only outcome this readout can establish: "no gap larger than X at High even where JOINT trained on the evaluated configuration and ISO never saw a composed one." Quoted as an exclusion with X from the grid's own seed dispersion, never as an absence.*

The same entry states that the readout "is NOT a held-out test", that a positive Γ_H is "NOT separable from in-distribution advantage", and that a negative one is "exploratory".

7. The T = 2000 subsample (same ruling):

> ***What it may be cited for:** whether the differential drift measured at T = 1000 (review §6 item 1: 0.01209 per 100 updates) visibly persists, closes or reverses. **What it may not be cited for:** any claim that Γ "holds" or "vanishes" at T = 2000.*

8. The mediation void rule (the frozen analysis plan, 2026-08-02):

> ***If realized dose does not vary with assigned p, the dose figure has no x-axis and is declared VOID** — registered now, not decided after seeing the first stage.*

9. The pre-unblind rulings (2026-09-26):

#	question	RULED
(i)	what "−" means in the branch table	**Rejection with negative sign** (|z| > z_α, Γ < 0). A negative estimate that does not reject is a null. Ratified as built.
(ii)	the cell (completion +, survival null), absent from the registered table	**REGISTERED NOW as branch A_c — the founding claim on the founding primary.** […]
(iii)	test statistic	**Normal z, z_α = 2.2365**, as the frozen plan wrote; the t critical value is printed, not applied. Ratified as built.
(iv)	Γ(t) "sign stable"	**All 11 point estimates, t = 500 … 1000, share one nonzero sign**, per co-primary; t = 1000 is the grid's own eval. Consequence accepted in advance: on a null Γ, chance flips read UNSTABLE. Ratified as built.

#	question	RULED
(v)	Γ_4 sample and threshold	**JOINT k = 72 vs ISO-4 k = 20**, SE `sqrt(s_J²/72 + s_4²/20)`, "rejects" and "CI includes 0" at z_α = 2.2365. (Γ null, Γ_4 rejects positive) stays UNREGISTERED. Ratified as built.
(vi)	Γ_H reading	**As (i), at z_α = 2.2365.** Ratified as built.
(vii)	primary k	**k = 72 stays primary**; the registered-ladder k = 60 prefix is reported beside it, UNDERPOWERED-flagged.
(viii)	re-evaluate the 144 T* = 1000 checkpoints on the post-unblind card	**YES.** […]

In (viii), "the post-unblind card" is GPU 3.

10. The secondary specification (2026-09-26, after unblinding and before the sweep was read):

> *(S1) PAVA + OLS slope CI; (S2) single-hinge breakpoint, UNDERPOWERED iff the CI spans {0.125 … 0.375}; (S3) plane fit, bound = 95 % CI of 0.5·γ_n; (S4) training co-activity over updates 1…1000, VOID iff the first-stage CI contains 0; (S5) the same-card replicate is ADDED, beside Γ_4, unable to change its qualifier; (S6) 95 % intervals, Γ_4' at z_α. […]*

The omitted words name the code version, which identifies the repository.

E The registered outcome branches

"Positive" and "null" mean rejection or non-rejection at θ* in the confirmatory family, and "−" means rejection with negative sign (Appendix D, item 9). Branches A–D were registered on 2026-08-13, each with a framing; the registered table's venue-lean column is omitted. A_c was added on 2026-09-26, before unblinding, for a cell the table lacked.

branch	completion	survival	headline claim, as registered (A_c: its registered claims, summarized)
A	+	+	founding registration, verbatim
B	null	+	composition is paid in **agents**, not task return
C	null	null	the instrument, and what it excludes
D	− on either co-primary	− on either co-primary	not separable from the matched-budget asymmetry
A_c	+	null	the founding claim on the founding primary; the survival null quoted as an exclusion; no asymmetry claim

The result is branch B, with the inert-share qualifier attached (§9.2).

F Grid provenance

The grid was 252 runs from 2026-09-09 to 2026-09-22. Seeds 1–12 of all ten arms (ISO, JOINT and eight secondary arms) ran on an RTX PRO 6000, then seeds 13–20 of all ten arms and 21–46 of ISO and JOINT on GPU 1. The confirmatory artifact holds all 72 seeds of ISO and JOINT on GPU 1: 76 runs executed there and seeds 13–46 copied from the grid. The superseded RTX PRO 6000 seeds

enter only a descriptive diagnostic of the GPU effect. Seeds 1–12 of the secondary arms were re-run on GPU 1 (96 runs), so every analysed secondary run is on GPU 1 and Γ_4' is a same-GPU contrast. ISO-4 and the T = 2000 subsample trained on GPU 2, so Γ_4 and $\hat{B}$ are across GPUs, and the post-unblind evaluations ran on GPU 3.

All arms share seed integers, and seed 0, the floor seed, never enters the grid. Runs are ordered seed-major and chunks end on seed boundaries, so an interruption leaves a smaller balanced design and both arms of a seed share one GPU. Every checkpoint archive's recorded sha256 matched on re-verification on 2026-09-27, 615 of 615. A partial re-measurement of floors on another rented GPU on 2026-08-14 is provenance only.

The grid runner computes no cross-arm quantity, which a unit test asserts along with the seed properties above, and the floor reports print dispersions only. The unblinding script refuses an unflagged artifact or a modified code version, later analysis scripts refuse to run before its log exists, and each log records one run of its stage.

G Gamma over the final half of training

Sign unstable over the final half, on both co-primaries: neither meets the registered rule that all 11 point estimates share one sign (Appendix D, items 2 and 9). Values are rates, and the Šidák-corrected intervals are registered at t = 1000 and descriptive before it. Checkpoints before t = 1000 were evaluated on GPU 3; t = 1000 is the grid's own evaluation, and re-evaluating it on GPU 3 leaves every sign unchanged (cross-GPU agreement: §11).

Completion, k = 72 per arm:

t	Γ(t)	sd(Γ)	Šidák CI	sign
500	+0.0017	0.00321	[−0.0055, +0.0089]	+
550	−0.0017	0.00353	[−0.0096, +0.0062]	−
600	+0.0018	0.00362	[−0.0063, +0.0099]	+
650	+0.0015	0.00481	[−0.0093, +0.0122]	+
700	−0.0037	0.00470	[−0.0142, +0.0068]	−
750	−0.0147	0.00610	[−0.0283, −0.0010]	−
800	−0.0035	0.00677	[−0.0187, +0.0116]	−
850	−0.0053	0.00729	[−0.0216, +0.0110]	−
900	−0.0039	0.00843	[−0.0228, +0.0149]	−
950	−0.0016	0.01021	[−0.0245, +0.0212]	−
1000	−0.0173	0.01068	[−0.0412, +0.0066]	−

Survival rate, k = 72 per arm:

t	Γ(t)	sd(Γ)	Šidák CI	sign
500	+0.0022	0.00228	[−0.0029, +0.0073]	+
550	−0.0037	0.00220	[−0.0086, +0.0013]	−
600	−0.0027	0.00232	[−0.0078, +0.0025]	−
650	+0.0011	0.00242	[−0.0043, +0.0065]	+
700	+0.0031	0.00235	[−0.0021, +0.0084]	+
750	+0.0036	0.00255	[−0.0021, +0.0093]	+
800	+0.0020	0.00232	[−0.0032, +0.0072]	+
850	+0.0018	0.00234	[−0.0034, +0.0071]	+
900	+0.0056	0.00231	[+0.0004, +0.0108]	+
950	+0.0083	0.00219	[+0.0034, +0.0132]	+
1000	+0.0096	0.00216	[+0.0047, +0.0144]	+

H The secondary design

Everything here is secondary, with 95 % intervals bootstrapped over seeds within arm (B = 2000), conditional on the common evaluation draw and not family-corrected. Each arm has k = 20 seeds on GPU 1 with δ off. On the triangle $\nu = 1 - 2\mu + p$, and the arm names are the labels of Figure 13's table view.

arm	per-coupling frequency μ	co-occurrence p	fire-only share ν	survival mean	com mea
`sweep_c50_p000`	0.5	0	0	0.9241	0.78
`sweep_c50_p125`	0.5	0.125	0.125	0.9249	0.77
`sweep_c50_p250`	0.5	0.25	0.25	0.9261	0.77
`sweep_c50_p375`	0.5	0.375	0.375	0.9264	0.76
`sweep_c50_p500`	0.5	0.5	0.5	0.9289	0.77
`ident_c40_p000`	0.4	0	0.2	—	—
`ident_c40_p200`	0.4	0.2	0.4	—	—
`ident_c40_p400`	0.4	0.4	0.6	—	—
ISO-4, for comparison (GPU 2)	0.5	0	0	0.9256	0.77

The p = 0 sweep arm carries exactly ISO-4's training mixture.

estimate (points)	survival	completion
sweep slope per unit p (labelled "dose slope"; separate fit)	+0.89 [−0.33, +2.11]	−2.51 [−9.43, +4.12]
sweep endpoint, p = 0.5 minus p = 0	+0.48 [−0.21, +1.17]	−0.97 [−4.82, +2.87]
γ_p, the co-occurrence slope, per unit p	+3.26 [+1.37, +5.02]	+2.69 [−6.75, +12.45]
γ_n, the fire-only-share slope, per unit v	−2.37 [−4.19, −0.57]	−6.52 [−15.34, +2.42]
bound, 0.5 γ_n	−1.19 [−2.10, −0.28]	−3.26 [−7.67, +1.21]
change point (labelled "knee")	0.375 [0.125, 0.375], UNDERPOWERED	0.375 [0.125, 0.375], UNDERPOWERED
plane prediction of JOINT − ISO-4 ($\gamma_p \times 1$)	§9.2	+2.69; γ_p's interval contains both measured values

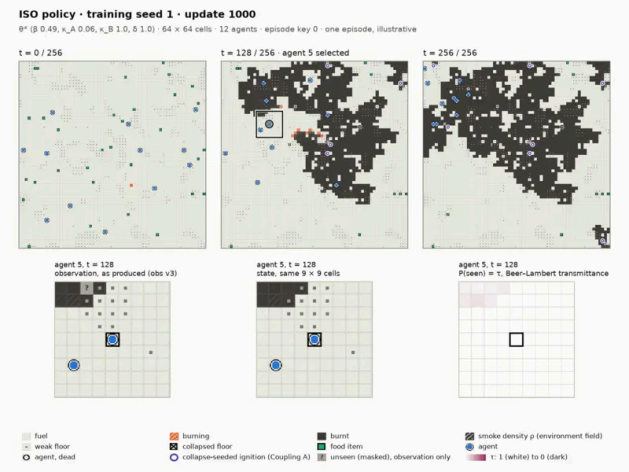
The plane $y = \gamma_0 + \gamma_p p + \gamma_n \nu$ is a seed-level ordinary least squares (OLS) fit over all eight arms, and the bound, the fire-only share's part of the sweep's endpoint change, excludes zero on survival. The sweep slope is a separate seed-level OLS fit over the five sweep arms, with an isotonic fit (PAVA) giving its monotone shape. The plane's slope along the sweep is $\gamma_p + \gamma_n$, so the two need not agree, and on completion they do not. The change point is the breakpoint of the best single-hinge fit. Linearity is assumed and checked only coarsely: arm-mean residuals lie within ±0.4 SE on survival and ±1.1 SE on completion, 8 points for 3 parameters.

Per-coupling frequency, co-occurrence and active share cannot all be held fixed on the triangle. Floors exist only for the sweep's p = 0.5 arm: the other sweep arms assume that floor, and the identification arms are ungraded. The counter's realized count is endogenous, so nothing here is its causal effect, and mediation is VOID (§9.2).

I Trained-policy episodes

Figures I1 and I2 show one episode of each confirmatory arm, separately, chosen by a rule fixed before any render: θ*, episode key 0, stochastic actions, training seed 1 at update 1000. Each opens at step 128 of 256. They illustrate the environment under a trained policy and grade nothing.

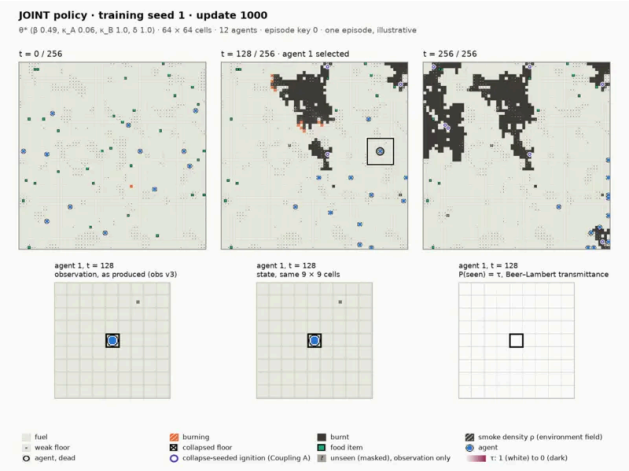
Figure I1. One episode of the ISO policy at θ*. The map is environment state, the smoke field ρ included. Selecting an agent shows its observation as produced (labelled obs v3: the masking observation, §3.4), the true cells, and the probability τ that each was seen. It cannot compare training protocols. Status: illustrative · grades nothing · Interact: play, step, click an agent.



Static companion of Figure I1 for the printed PDF.

Figure I2 shows the JOINT policy's episode under the same rule. One episode of one seed cannot carry a contrast that §9.2 measures over 72 seeds per arm and 512 evaluation episodes.

Figure I2. One episode of the JOINT policy at θ*, with the layers and panels of Figure I1; only the observation panel shows what the selected agent perceives. It cannot compare training protocols. Status: illustrative · grades nothing · Interact: as Figure I1.



Static companion of Figure I2 for the printed PDF.

Footnotes

1. We used the LLM assistant Claude as a tool: under our rulings it implemented the environment, training and analysis code, built the figures, helped search the literature, verified the citations, and drafted this text from our recorded rulings and results. We checked every word and are responsible for it. [↩]

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